Computers and Education: Artificial Intelligence 10 (2026) 100594

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

From ethical principles to executable governance: A policy-as-code framework for trustworthy AI in higher education

Edmund Evangelista [a,*], Syed M. Salman Bukhari [b]

[a] College of Technological Innovation, Zayed University, Abu Dhabi, United Arab Emirates
[b] Department of Electrical Engineering, Capital University of Science and Technology, Islamabad, Pakistan

ARTICLE INFO

Keywords:
AI governance
Policy-as-Code
Responsible AI
Algorithmic fairness
Explainable AI
Higher education
Learning analytics

ABSTRACT

Artificial intelligence holds great potential to transform higher education, but a persistent gap remains between ethical aspirations and their practical, auditable enforcement. This study addresses that gap by developing and validating an end-to-end executable governance framework grounded in a policy-as-code (PaC) paradigm. Using student dropout prediction as a high-stakes example, the framework operationalizes governance through an automated gatekeeper, a multi-strategy fairness mitigation toolbox, and a tamper-evident audit chain for full reproducibility. The governance compliance was tested across sixteen fixed model configurations evaluated under five policy tiers (strict, medium, lenient, and two deployment-realistic variants). None were approved, as fairness violations, dominated by socio-economic (imd_band), educational (highest_education), and intersectional attributes, accounted for majority of all block reasons, even under relaxed thresholds. The mitigation strategies such as ExponentiatedGradient and ThresholdOptimizer improved subgroup equity but induced severe performance or calibration regressions (AUC ↓ 0.92 → 0.78, ECE ↑ 0.04 → 0.23, Recall ↓ ≈ 0.17). In comparison, explainability coverage (100%) and 95th percentile governance decision latency (P95 ≈ 23.5 ms) confirmed negligible operational overhead. The findings identify fairness and calibration, not computational cost, as the binding limitations for responsible deployment. By integrating human-in-the-loop policy formation and diagnostic auditability into the AI lifecycle, this study provides a reproducible blueprint for institutions to move from static ethical principles to participatory, verifiably trustworthy AI systems.

1. Introduction

Artificial intelligence (AI) is increasingly integrated in higher education through applications such as personalized learning, predictive analytics, adaptive assessment, and automated student support (Baker & Siemens, 2012; Holmes et al., 2019). While these systems offer substantial benefits, they also raise governance challenges that directly affect institutional trust and accountability. The non-representative data can propagate biases (Dwork et al., 2012; Kearns et al., 2018), black box model behaviour affects transparency (Ribeiro et al., 2016; Selwyn, 2019), weak auditing limits accountability (Slade & Prinsloo, 2016; Williamson & Selwyn, 2019), and the absence of uncertainty quantification reduces confidence in operational reliability. Although recent research emphasizes fairness, transparency, and inclusivity (Holmes et al., 2021; Kearns et al., 2018; Plečko & Bareinboim, 2024), most approaches remain descriptive rather than enforceable. In practice, institutions require governance mechanisms that are executable, testable, and auditable within the modelling pipeline itself, rather than aspirational guidelines applied post hoc.

This study meets the need by defining institutional governance as policy-as-code (PaC), encoding requirements as machine-readable thresholds enforced by a gatekeeper during model evaluation and deployment. Using the Open University Learning Analytics Dataset (OULAD) (Kuzilek et al., 2017) for dropout-risk prediction, a representative, high-stakes educational task, we implement and evaluate an end-to-end governance pipeline that produces verifiable, policy-linked evidence for every experimental run. The framework evolves through two stages, reflecting a shift from static assurance toward diagnostic governance.

- **Stage 1 (Prototype)** translates institutional requirements into fixed threshold policies enforced by a binary gatekeeper (APPROVE/BLOCK). While suitable for basic assurance, this design proved overly rigid: under the initial thresholds and a single post-processing

* Corresponding author. Abu Dhabi Campus Zayed University, P.O. Box 144534, Abu Dhabi, United Arab Emirates.
E-mail address: Edmund.evangelista@zu.ac.ae (E. Evangelista).

https://doi.org/10.1016/j.caeai.2026.100594
Received 14 November 2025; Received in revised form 29 March 2026; Accepted 8 April 2026
Available online 8 April 2026

mitigation, none of the evaluated configurations satisfied all governance constraints.

- **Stage 2 (Advanced Framework)** extends this prototype into a diagnostic, uncertainty-aware system through several enhancements: (i) a time-respecting FIT-CAL-TEST evaluation protocol to prevent temporal leakage and support calibration validity; (ii) a multi-strategy fairness toolbox combining post-processing (ThresholdOptimizer) and in-processing (ExponentiatedGradient) mitigations; (iii) bootstrapped 95% confidence intervals for all major metrics; (iv) realistic multi-tier policy sets (strict, medium, lenient); (v) diagnostic gatekeeping that quantifies per-metric shortfalls and identifies binding constraints; and (vi) tamper-evident auditability through a hash-chained (SHA-256) governance log covering pre-processing, modelling, explanation, and enforcement stages.

These enhancements transform governance from a static checklist into a data-driven diagnostic system. The previous research in responsible educational AI has advanced documentation practices, fairness metrics and mitigations, ethical guidelines, and audit concepts, but largely in isolation. The documentation frameworks emphasize transparency without enforcement; fairness toolkits provide metrics and mitigations without institutional deployment semantics; governance frameworks remain non-executable; and audit mechanisms are often conceptual or detached from live decision pipelines. The significance of this study lies in effectively integrating existing fairness metrics and learning algorithms into a practical governance framework, rather than introducing new ones. By coupling a PaC schema with a confidence interval (CI)-aware automated gatekeeper, multi-strategy mitigation evaluation, and a tamper-evident audit chain, the framework enables deterministic, diagnosable, and reproducible governance decisions. This integration is particularly salient in higher education, where algorithmic decisions are high stakes, delayed, and subject to institutional review rather than real-time automation. A detailed comparison with previous research is provided in Tables 1 and 2.

1.1. Motivation and problem statement

Educational AI systems increasingly influence high-stakes outcomes such as admissions, grading, advising, and resource allocation. In these contexts, aspirational ethics statements are insufficient (Holmes et al., 2019; Williamson & Selwyn, 2019). The institutions require operational mechanisms that can: (i) encode governance expectations precisely; (ii) evaluate models under time-respecting, leakage-free protocols; (iii) enforce policy decisions consistently and transparently; (iv) diagnose failures and trade-offs to guide remediation; and (v) audit the entire process through tamper-evident, reproducible records (Mökander et al., 2022; Raji et al., 2020). A few existing frameworks satisfy all five requirements simultaneously (Holmes et al., 2021).

To address this gap, we implement a PaC governance framework for dropout-risk prediction on the OULAD dataset. The policies establish minimum acceptable standards for performance (Area Under the Curve-AUC, macro-F1, and positive class recall), calibration (Expected Calibration Error-ECE and Brier score), fairness (Demographic Parity-DP, Equalized Odds-EO, and worst-case intersectional gaps), explainability (minimum SHapley Additive exPlanations-SHAP coverage), and operations (95th Percentile-P95) latencies for explanations and governance checks). A gatekeeper evaluates each candidate configuration against these thresholds using a conservative, CI-aware decision mode and records outcomes in a hash-chained audit trail for reproducibility.

Fig. 1 positions this motivation within the larger landscape of educational AI, highlighting that although these systems provide benefits such as personalization, efficiency, and improved accessibility, they also present systemic challenges, including potential bias, limited transparency, and operational fragility.

Table 1
Comparison of Previous Studies vs. Proposed Framework.

Focus Area	Previous Studies	Limitations	Our Framework
Fairness	Bias audits for subgroup fairness (Dwork et al., 2012; Kearns et al., 2018); causal fairness analysis (Plečko & Bareinboim, 2024). Methods include adversarial debiasing.	Post-hoc analysis only; no enforcement	Multi-strategy fairness mitigation + policy gatekeeper
Explainability	SHAP/LIME interpretability in student models (Lundberg & Lee, 2017; Ribeiro et al., 2016).	Explanations not tied to governance	SHAP integrated into compliance schema (FIT–CAL–TEST)
Governance	Ethical guidelines and principle-based codes (United Nations Interregional Crime and Justice Research Institute, 2024; Williamson & Piattoeva, 2019)	Non-operational, static	PaC schema across AI lifecycle
Auditability	Blockchain proposals for credentialing and record-keeping (Yli-Huumo et al., 2016)	Conceptual; no real-time integration	SHA-256 blockchain-style immutable audit logs
Deployment	Analyses of weak alignment with workflows and ethical dilemmas in practice (Slade & Prinsloo, 2016; Williamson & Selwyn, 2019)	Weak alignment with workflows	Enforceable governance gatekeeper for student-facing apps

Table 2
Comparison of related governance approaches and the proposed framework.

Previous research	Typical Capabilities	Limitations for Institutional Governance	Our contribution
Documentation frameworks (Model Cards, Datasheets)	Structured reporting of model and data properties	Descriptive only; no automated enforcement; no deployment decision logic	Machine-executable policies with automated approval/blocking and diagnostic outputs
Algorithmic auditing pipelines	Retrospective fairness and performance audits	Often manual, post-hoc, and detached from deployment workflows	CI-aware gatekeeper that evaluates compliance during model evaluation and deployment
Fairness toolkits (metrics + mitigations)	Measurement and mitigation methods (e.g., DP, EO, EG, TO)	No institutional policy tiers; no notion of deployment eligibility	Policy-tiered governance where mitigations are evaluated as candidates under executable constraints
General PaC engines (DevSecOps)	Enforce software and security policies	Not ML-native; no fairness, calibration, or explainability semantics	PaC schema customized to ML governance (performance, calibration, fairness, explainability, operations)
Audit logging/ provenance systems	Tamper-evident logs or traceability	Not integrated with governance decisions or model approval	Hash-chained audit ledger binding policies, metrics, explanations, and gatekeeper decisions

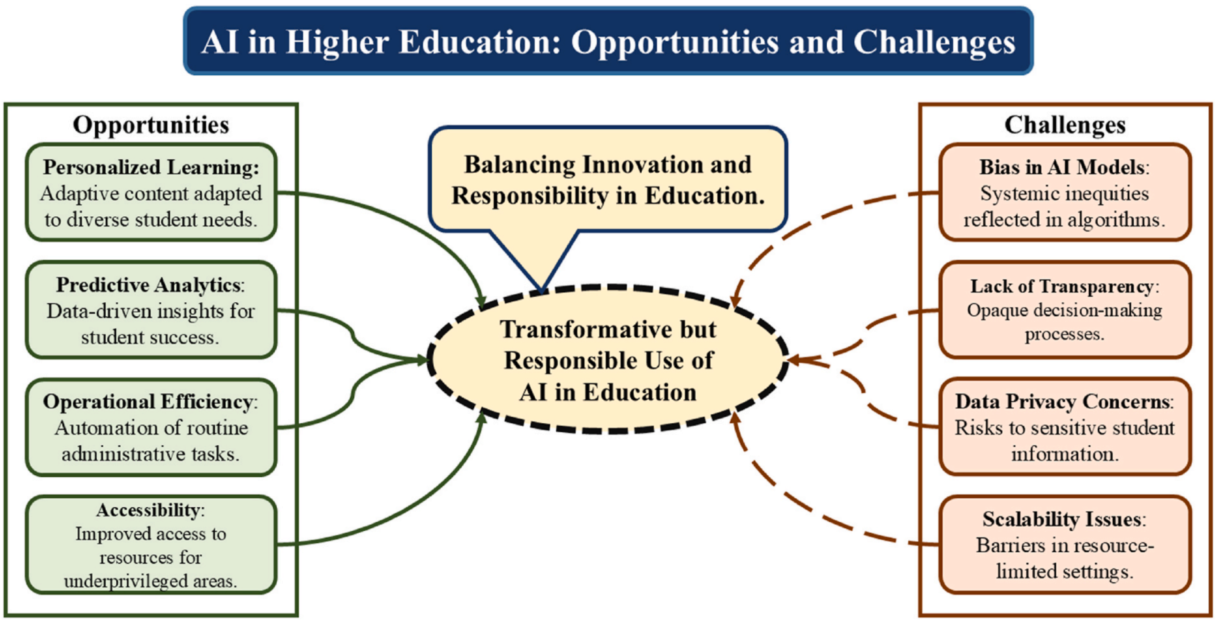

Fig. 1. Strategic opportunities and governance challenges in educational AI systems.

1.2. Key contributions

This study advances responsible educational AI by converting ethical principles into executable, auditable, and diagnostically interpretable governance mechanisms. The proposed PaC framework establishes a reproducible pipeline that quantifies compliance, uncertainty, and accountability. Specifically, we contribute:

- **Executable Policy-as-Code Schema:** A machine-readable specification encoding institutional requirements for performance, calibration, fairness (including intersectional analysis), explainability coverage, and operational service levels. Each policy is cryptographically fingerprinted (SHA-256) and automatically enforced, enabling transparent and repeatable decision logic.
- **Diagnostic Gatekeeping Engine:** An enforcement module that extends binary approval into a diagnostic decision process. It quantifies per-metric shortfalls relative to policy thresholds, identifies binding constraints, and generates policy-tuning suggestions through three artifacts: metrics with 95% CIs, shortfall heatmaps, and policy-tuning recommendations.
- **Uncertainty-Aware Evaluation Design:** A time-respecting FIT-CAL-TEST protocol combined with bootstrap CIs for all major metrics, ensuring statistically robust model evaluation and governance decisions.
- **Multi-Strategy Fairness Mitigation Analysis:** A comparative evaluation of post-processing (ThresholdOptimizer) and in-processing (ExponentiatedGradient) methods, revealing how mitigation strategies clarify, rather than eliminate, the trade-off surface between performance and fairness.
- **Tamper-Evident Auditability:** A hash-chained governance log recording pipeline events, configurations, metrics, explanations, policy evaluations, and artifact hashes, providing verifiable accountability with negligible operational overhead (95th percentile) governance decision latency $\approx$ 23.5 ms; SHAP coverage $\approx$ 100%).

While validated with the OULAD dataset, the framework can be used in other educational contexts that require practical algorithmic accountability. Based on this goal, we address these research questions:

- **RQ1:** How can institutional AI governance policies be encoded as executable, auditable rules?
- **RQ2:** Under a time-respecting evaluation, how do performance, calibration, and fairness interact, and how should institutions interpret these trade-offs under uncertainty?
- **RQ3:** Does using a multi-strategy mitigation toolbox improve compliance feasibility compared to a single mitigation approach?
- **RQ4:** What operational overheads and explainability coverage levels are introduced by policy enforcement?
- **RQ5:** How do diagnostic artifacts and CI-aware evaluations inform policy refinement and responsible adoption decisions?

The remainder of this paper is structured as follows. Section 2 reviews related research studies in educational AI governance, fairness, calibration, and auditability. Section 3 details the methods, including the policy schema, fairness mitigations, evaluation protocol, and governance architecture. Section 4 presents result for the advanced pipeline, emphasizing feasibility, uncertainty, and diagnostic insights. Section 5 discusses implications for institutional adoption, limitations, and future research directions, and Section 6 concludes the article.

2. Related work

Research in AI for higher education has produced increasingly accurate predictive models for personalized learning, retention analysis, and student success analytics (Baker & Siemens, 2012; Macfadyen & Dawson, 2010). Recent advances have also expanded the role of AI in educational recommendation systems, where intelligent models assist learners in identifying suitable courses, learning resources, and academic directions. For example, recent studies such as LE-DLCM (Ma et al., 2025) proposes a large-language-model (LLM)-based framework that decouples learner and course representations to enhance course recommendation through enriched semantic modelling of learner profiles and educational content. Such approaches show the growing integration of advanced machine learning and generative AI techniques in smart education environments to improve personalization and adaptive learning experiences.

As educational AI systems evolve, research is increasingly stressing the need for fairness, transparency, and accountability. As predictive and recommendation systems increasingly influence academic decision-making, concerns have emerged regarding bias propagation, limited interpretability, and weak institutional oversight. However, despite this progress, a persistent gap remains between responsible AI principles and their operational enforcement in institutional settings. Much of the existing literature clarifies what ethical AI should achieve but provides limited guidance on how such assurances can be implemented, evaluated, and audited as part of routine model development and deployment. To address this gap, the following section reviews previous research across three interrelated areas, analytical tools, governance frameworks, and auditability mechanisms, and situates the contribution of this study relative to their limitations.

2.1. Analytical tools for fairness and explainability

A substantial body of research has developed analytical tools for examining model behaviour after training. In algorithmic fairness, formal definitions and metrics such as DP and EO enable systematic assessment of subgroup biases (Dwork et al., 2012; Kearns et al., 2018). Similarly, interpretability methods including Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) have been widely adopted to explain individual predictions and feature contributions in student-facing models. While these tools are essential for diagnosing bias, they typically operate post hoc, identifying issues after models have already been trained. As a result, fairness and explainability are often treated as optional evaluation steps rather than enforceable institutional requirements. Moreover, most studies report point estimates without quantifying statistical uncertainty, and explanation outputs are rarely tied to explicit governance thresholds. As a result, these approaches can answer analytical questions such as "Is the model fair?" or "Which features matter?" but they do not support institutional decisions such as "Does this model meet policy, and can it be deployed?"

2.2. Governance frameworks and the challenge of enforcement

Beyond analytical tooling, a second research stream has articulated ethical and governance frameworks for AI in education (Holmes et al., 2021; United Nations Interregional Crime and Justice Research Institute, 2024; Williamson & Piattoeva, 2019). These efforts emphasize principles such as fairness, transparency, accountability, and human oversight, and they have informed institutional guidelines and emerging regulatory discussions. However, such frameworks are largely descriptive and non-executable, functioning as codes of conduct rather than machine-enforceable specifications. This limitation is increasingly consequential. The regulatory initiatives such as the EU AI Act (Smuha, 2025) and the NIST AI Risk Management Framework (National Institute of Standards and Technology, 2023) emphasize the need for auditable evidence of compliance, rather than narrative assurances. In practice, institutions face accountability gaps (Williamson & Piattoeva, 2019), misalignment between governance policies and technical workflows (Williamson & Selwyn, 2019), and diminished trust when algorithmic decisions cannot be consistently justified (Selwyn, 2019). Addressing these challenges requires governance mechanisms that are computable, verifiable, and continuously enforceable throughout the AI lifecycle.

2.3. Auditability and provenance in AI systems

Auditability is widely recognized as a cornerstone of responsible AI, but many proposed mechanisms remain conceptual or detached from live system operation. The previous research has explored immutable record-keeping and blockchain-based credentialing in educational contexts (Yli-Huumo et al., 2016), introducing important ideas around tamper resistance and traceability. However, these approaches are rarely integrated with AI model evaluation, policy enforcement, or deployment decisions. Without real-time linkage between governance decisions and verifiable records, institutions risk "audit-washing", asserting compliance without cryptographic proof (Mökander et al., 2022; Raji et al., 2020). The effective accountability requires systems that record governance actions as they occur, bind decisions to concrete evidence, and preserve provenance across preprocessing, modelling, evaluation, and explanation stages.

The literature reveals three persistent limitations: (1) analytical tools remain retrospective and lack preventive enforcement capabilities; (2) governance frameworks are largely descriptive rather than actionable; and (3) auditability mechanisms are conceptual or loosely coupled to AI workflows. This study proposes a diagnostic governance pipeline that integrates institutional policy into model evaluation and deployment. The framework operationalizes governance through a combination of executable policies, uncertainty-aware decision logic, mitigation analysis, and tamper-evident provenance.

To clarify this contribution, Table 1 compares previous educational AI studies with the proposed framework across key focus areas such as fairness, explainability, governance, auditability, and deployment alignment. This comparison highlights how earlier research typically addresses individual aspects in isolation, whereas the proposed approach integrates them into a unified enforcement workflow.

Table 2 complements this comparison by situating the framework relative to broader categories of governance-related systems, including documentation frameworks, algorithmic auditing pipelines, fairness toolkits, general PaC engines, and audit logging mechanisms.

Tables 1 and 2 show that the novelty of this study lies not in introducing new fairness metrics or mitigation algorithms, but in combining existing components into a single executable governance loop that supports deterministic approval decisions, diagnostic feedback, and verifiable accountability. By moving from principles to practice, the proposed framework provides an empirical demonstration, within higher education analytics, of how responsible AI governance can be executed, quantified, and audited as part of routine model development, rather than treated as a post-hoc or purely procedural concern.

3. Methods and framework design

This study designs and validates an executable, policy-driven governance pipeline for educational dropout-prediction models using the OULAD dataset. Fig. 2 presents a high-level conceptual overview of the governance architecture, showing how institutional requirements are formalized as PaC, deterministically evaluated by a gatekeeper, optionally trigger fairness mitigation, and are immutably recorded in a tamper-evident audit chain. This approach highlights governance, reproducibility, fairness auditing, calibration, explainability, and uncertainty awareness, ensuring every stage is transparent, auditable, and enforceable by code. The advanced design introduces bootstrap-based CIs for all metrics, conservative policy decisions based on CI bounds, diagnostic reporting tables, and hash-chained audit logging, resulting in a deployment-ready governance framework.

Fig. 3 puts this conceptual design into practice by showing the complete execution process through two integrated layers: the Data & Modeling Pipeline (on the top) and the Governance, Diagnostics & Audit layer (on the bottom), all managed by PaC (Box 0). The modeling pipeline follows a chronological FIT→CAL→TEST sequence, including policy-driven preprocessing, model training, calibration, fairness mitigation, and initial evaluation against performance, calibration, and fairness thresholds (Boxes 1-6). The governance layer then performs SHAP-based explainability checks (Box 7), detailed diagnostics with metrics reported alongside 95% bootstrap CIs and policy shortfall analysis (Box 8), and operational governance checks such as decision latency (Box 9 All significant artifacts and events are immutably recorded through a hash-chained audit trail (Box 11). Finally, the Gatekeeper (Box 10) enforces the proof-of-concept (PoC) rules to approve or block configurations, logging explicit reasons and producing governance outputs (e.g., tuning suggestions from Box 8) that enable a structured feedback loop for potential refinement while preserving

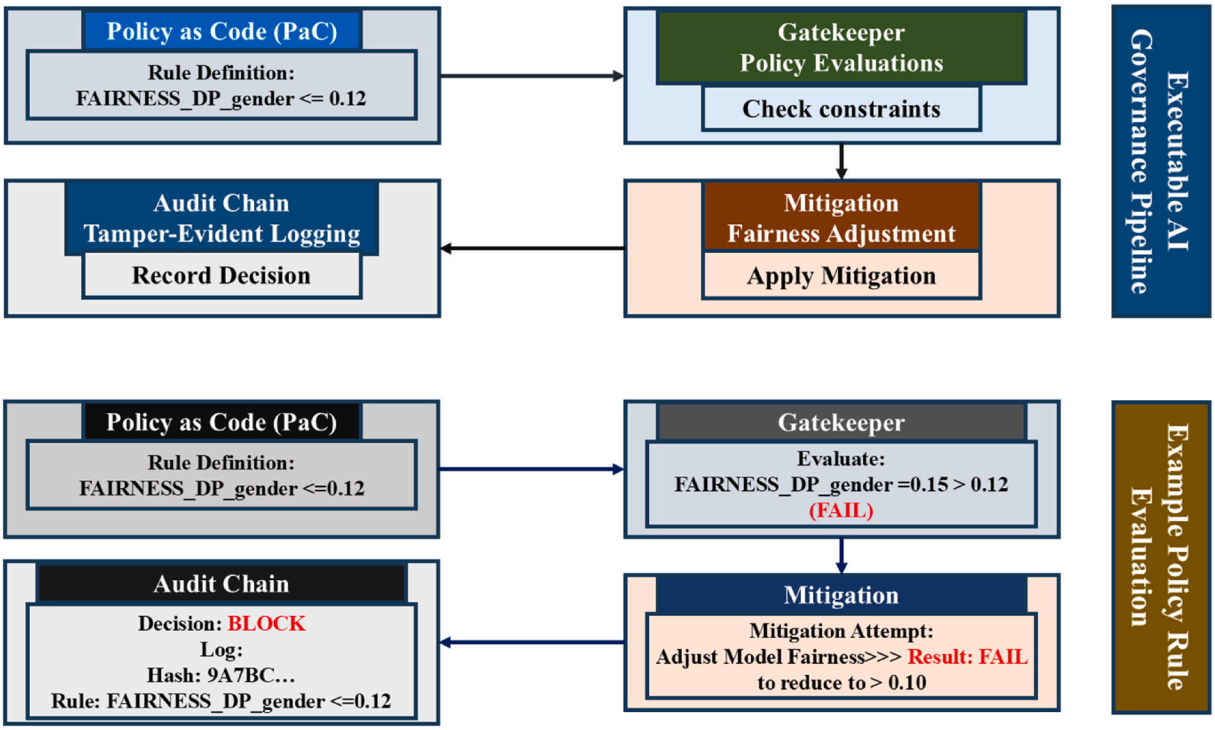

Fig. 2. Executable AI governance pipeline with end-to-end policy evaluation.

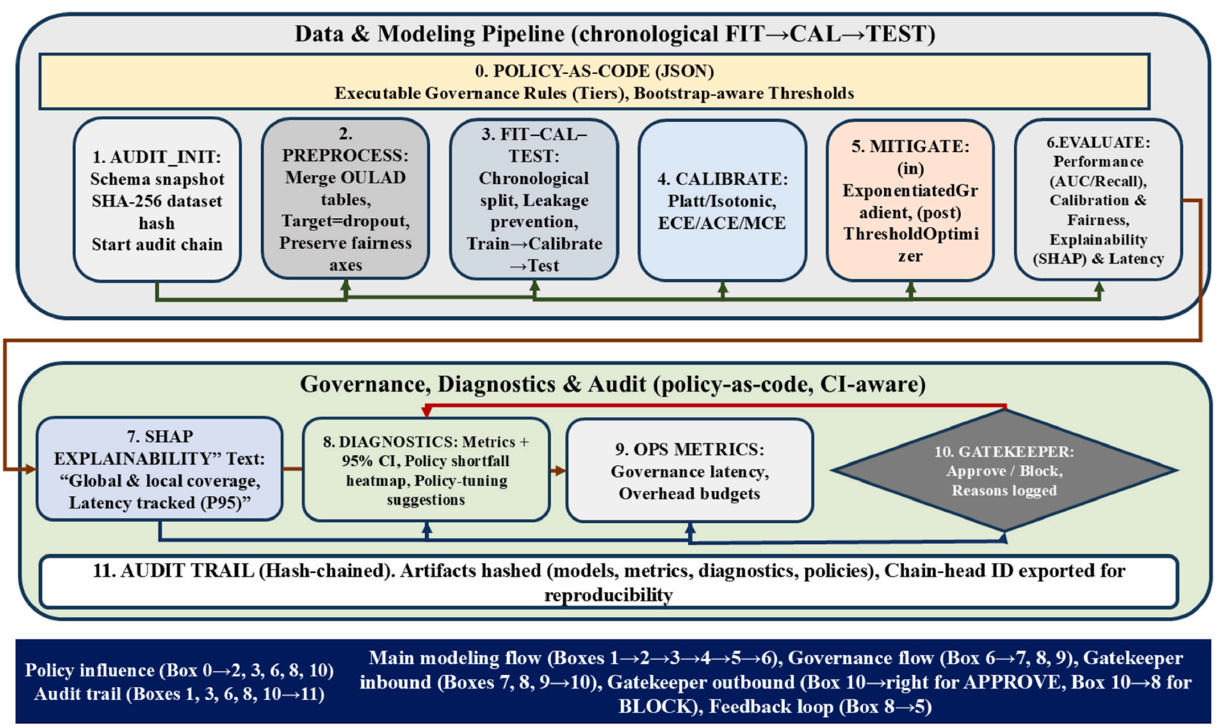

Fig. 3. End-to-end governance pipeline for educational AI models.

reproducible accountability.

3.1. Policy-as-Code Schema

To ensure that governance is transparent, reproducible, and technically enforceable, all evaluation and deployment thresholds were formalized as machine-readable policy files. This PaC design encodes institutional expectations across performance, calibration, fairness, explainability, and operational dimensions into structured JSON objects. Each policy file is cryptographically fingerprinted (SHA-256) and linked to subsequent audit-log entries, guaranteeing one-to-one traceability between policy definitions and their recorded governance outcomes. All metrics are evaluated with bootstrapped 95% CIs, and the gatekeeper applies a conservative decision mode, using lower CI bounds for "≥" metrics and upper bounds for "≤" metrics, to ensure that approvals are statistically robust rather than point-estimate dependent.

- **Performance metrics:** Minimum AUC, macro-F1, dropout-recall thresholds capture ranking quality, balanced accuracy, and sensitivity to at-risk learners, the group of highest institutional concern.
- **Calibration metrics:** Brier Score, Expected Calibration Error, Adaptive Calibration Error, and Maximum Calibration Error assess both average and worst-case reliability of predicted dropout probabilities.
- **Fairness metrics:** Policies enforce DP and EO across gender, imd_band, and highest_education, as well as their intersection (gender × imd_band). Each subgroup is required to meet $n \geq 30$ for statistical stability.
- **Explainability metrics:** SHAP coverage (fraction of test examples with valid explanations) and 95th percentile latency for single-instance explanations is monitored to maintain interpretability under operational constraints.
- **Operational governance metrics:** 95th percentile governance event latency and audit overhead measure the runtime cost of policy evaluation and logging, ensuring that governance remains deployable in practice.
- **Metadata:** Each policy file records its schema version, timestamp, integrity hash, fairness axes, calibration settings, and artifact paths, forming the root of the hash-chained governance ledger.

Five policy tiers were defined, Strict, Medium, Lenient, and two deployment-realistic variants (Medium_Realistic and Lenient_Realistic), as shown in Table 3. The threshold values were determined using a hybrid empirical-normative approach. The initial ranges were informed by bootstrap confidence distributions of baseline models to ensure feasibility under realistic deployment conditions and were then aligned with commonly cited governance tolerances in the responsible AI literature and regulatory guidance. The resulting tiers represent progressively relaxed institutional risk postures rather than purely statistical artifacts. This PoC did not include any formal process to gather stakeholder input. Instead, the thresholds should be interpreted as governance proxies designed to show enforceability, diagnostic power, and auditability. In a real educational deployment, these thresholds would be co-developed through participatory governance processes involving program directors, instructors, learning-analytics specialists, and ethics or quality-assurance committees, who would deliberate on acceptable trade-offs between predictive reliability, fairness, and accountability.

Each policy JSON file integrates a qualitative rationale for every threshold, linking each quantitative limit to its institutional purpose (e.g., safety priority: minimizing false negatives for at-risk learners). These explanations are recorded through the auditable THRESHOLD_SOURCES schema within the GovernanceConfig, ensuring that every metric is not only numerically defined but contextually justified within an educational governance setting. All fairness thresholds are defined in terms of absolute group-level biases. Specifically, the EO gap is computed as the maximum absolute difference in true positive rates (TPR) and false positive rates (FPR) between any protected subgroup and the reference group, while the DP gap is defined as the absolute difference in positive prediction rates. EO threshold of 0.12 corresponds to a maximum allowable bias of 12 percentage points.

This schema ensures that model approval is deterministic, uncertainty-aware, and reproducible. Any configuration that violates thresholds within or beyond its confidence bounds is automatically blocked, and the specific reasons are recorded in the immutable governance log. The combination of empirical threshold derivation, qualitative rationale embedding, and hash-chained traceability establishes a governance framework that is both technically rigorous and institutionally transparent.

Table 3
PaC thresholds for performance, calibration, fairness, explainability, and governance.

Metrics	Strict	Medium	Lenient	Medium (Realistic)	Lenient (Realistic)
AUC (min)	0.80	0.78	0.75	0.78	0.75
Dropout Recall (min)	0.75	0.70	0.65	0.70	0.65
Macro-F1 (min)	0.80	0.78	0.75	0.78	0.75
ECE (max)	0.05	0.07	0.10	0.12	0.18
Brier Score (max)	0.18	0.20	0.22	0.20	0.22
DP Gap (per axis, max)	0.03	0.05	0.07	0.10	0.12
EO Gap (per axis, max)	0.05	0.07	0.09	0.12	0.15
Intersectional Gap (max)	0.06	0.08	0.10	0.12	0.15
SHAP Coverage (min)	0.95	0.90	0.80	0.90	0.80
SHAP Latency P95 (ms, max)	250	400	600	400	600
Overhead P95 (ms, max)	150	250	350	250	350

3.2. Data & splits

This section provides a detailed description of the dataset used in this study, including its provenance, structure, class distribution, and preprocessing procedures. Because reproducibility and governance transparency are central to the proposed framework, all dataset preparation steps were designed to ensure traceability, leakage prevention, and reproducible evaluation. The following subsections describe (i) the dataset source and target construction, (ii) preprocessing and governance artifacts, and (iii) the time-respecting data split used for model training, calibration, and testing.

3.2.1. Dataset provenance and target construction

This study uses the OULAD (Kuzilek et al., 2017), a publicly available benchmark released by the UK Open University and widely adopted in educational data-mining research. The dataset aggregates student-level records across 22 modules and four course presentations (2013B, 2013J, 2014B, 2014J), covering academic years 2013-2014. In total, the merged dataset comprises 32,593 learners, providing a large and institutionally realistic population for evaluating governance mechanisms in high-stakes educational AI. OULAD integrates heterogeneous data sources, including:

- **Demographic and Socio-economic Attributes:** Gender, age band, disability status, highest previous education, and imd_band (Index of Multiple Deprivation band).
- **Academic Background:** Studied credits, previous attempts, and entry qualifications.
- **Behavioural Engagement Signals:** Fine-grained Virtual Learning Environment (VLE) activity logs, including interaction counts and temporal engagement markers.

The prediction target was derived from the 'final_result' field and operationalized as a binary dropout outcome:

- **Dropout (1):** Learners whose final status was Withdrawn or Fail.
- **Retained (0):** Learners whose final status was Pass or Distinction.

This definition reflects institutional priorities in early-warning systems, where failing to identify at-risk students represents the highest operational risk. In the full dataset, the resulting class distribution comprises 17,208 dropout cases (52.8%) and 15,385 retained cases (47.2%), producing a balanced and realistic supervision signal. For intersectional fairness analysis, subgroups were defined by the Cartesian product of gender and imd_band categories present in OULAD. Across the evaluated dataset, intersectional subgroup sizes ranged from approximately $n \approx 30$ to $n > 4000$ learners, with most groups exceeding several hundred samples. To ensure statistical reliability, fairness metrics were computed only for subgroups meeting a minimum sample size threshold.

3.2.2. Preprocessing and governance artifacts

The preprocessing was designed to satisfy two objectives simultaneously: preventing target leakage and ensuring auditable, governance-aligned data handling. All OULAD source tables were merged into a unified, student-level dataset. The outcome-linked fields (final_result, date_unregistration, and the derived target label) were excluded from the predictive feature set to eliminate leakage. An intersectional attribute (gender × imd_band) was generated dynamically during auditing to capture compounded inequities without exposing sensitive attributes to the predictive model. Where explicit consent indicators were absent in the original dataset, a deterministic consent simulation was applied using a fixed random seed, and the resulting exclusions were logged for ethical traceability. All preprocessing steps followed reproducible, audit-ready principles:

- Explicit categories, such as "?" in imd_band, were used for categorical anomalies instead of imputing them, so that missing data stays visible and can be reviewed.
- Engagement variables used zero values to represent the absence of activity, rather than statistical imputation.
- Nominal attributes were normalized for consistent casing, while numeric features were left unscaled to accommodate tree-based models.

Three immutable data artifacts were generated and recorded with SHA-256 hashes in the governance ledger: (1) a leakage-free dataset for model training and evaluation; (2) a full dataset retaining sensitive attributes for governance, fairness, and interpretability analysis; and (3) a schema snapshot documenting feature names, data types, and missing values. These artifacts establish a transparent preprocessing lineage that is reproducible and verifiable across runs.

3.2.3. Time-respecting splits and uncertainty estimation

To better simulate authentic deployment scenarios, the framework utilized a time-respecting FIT-CAL-TEST split instead of relying on random data partitioning. For each module, the most recent course presentation was reserved as the TEST set (future data), while earlier presentations were allocated to FIT (training) and CAL (calibration). The calibration sets were drawn from the most recent pre-test presentations to maintain temporal proximity without overlap. The calibration was implemented with cv = “prefit” to avoid cross-validation leakage. No k-fold or repeated cross-validation was used, as such procedures would violate temporal ordering across course presentations and risk information leakage in this longitudinal educational setting. To quantify uncertainty, nonparametric bootstrap resampling was applied to TEST set predictions to estimate 95% CIs for all key metrics, including predictive performance (AUC, recall, macro-F1), calibration error (ECE, ACE, MCE, Brier score), and subgroup fairness rates (DP, EO, and intersectional gaps). This design strengthens governance validity by ensuring that:

- **Metric Fidelity:** Reported metrics reflect temporally consistent deployment conditions.
- **Policy Credibility:** Compliance decisions are grounded in evidence representative of real-world operation rather than random-split optimism.
- **Diagnostic Continuity:** Uncertainty-aware outputs feed directly into analytic tables of the gatekeeper, enabling conservative enforcement and policy-tuning diagnostics.

The bootstrapping was selected over parametric standard errors or hypothesis testing to provide distribution-free uncertainty estimates suitable for conservative, policy-aware decision-making, rather than formal statistical inference. By supporting chronological separation and bootstrap-based uncertainty estimation, the pipeline evaluates governance under conditions that closely mirror live educational AI deployment, while explicitly accounting for metric variability in downstream policy enforcement.

3.3. Models & mitigation

This study compared multiple supervised learning models for dropout prediction and systematically integrated calibration, fairness mitigation, and governance diagnostics. The modeling pipeline was designed for reproducibility, comparability, and auditability across policy tiers. Each model configuration was stored with its parameters, preprocessing transformations, and governance lineage in the hash-chained audit ledger. The model set reflects two design imperatives: (i) robustness to heterogeneous of OULAD, mixed-type structure, and (ii) alignment with institutional governance requirements, where both predictive validity and subgroup equity are critical. In total, sixteen different model-policy configurations were evaluated, reflecting all combinations of model family, calibration strategy, fairness mitigation method, and protected attribute axis considered in this study.

3.3.1. Model families and preprocessing logic

Three complementary classifiers were implemented as baselines, chosen for their balance of interpretability, nonlinearity, and tabular performance:

- **Logistic Regression (LR):** Serves as a transparent linear baseline. Its coefficients are interpretable, training is efficient, and it supports in-processing fairness mitigation (ExponentiatedGradient).
- **Random Forest (RF):** An ensemble of bootstrap-aggregated decision trees, robust to noise and well suited to heterogeneous features.
- **Gradient Boosting Decision Trees (GBDT):** A stage-wise additive ensemble capable of capturing complex non-linear feature interactions, representing state-of-the-art performance for structured tabular data.

Each model was integrated into a reproducible pipeline with a fixed global random seed (SEED = 42) applied uniformly across all training, calibration, mitigation, and evaluation stages, and with version-controlled hyperparameters. The preprocessing followed a model-specific “ColumnTransformer” design. For LR, numeric features were standardized (z-score scaling), while categorical features were one-hot encoded with a minimum frequency threshold. For RF and GBDT, numeric features were standardized, and categorical variables were one-hot encoded using the same thresholding rule. The sensitive attributes (gender, imd_band, highest_education) were retained in full under the governance schema to support fairness auditing and subgroup policy enforcement. The class imbalance was mitigated using class_weight = “balanced” for LR and RF and equivalent weighting for GBDT, ensuring dropout (positive) cases were not underrepresented. As a result, this model set allows for fairness interventions to be applied both after and during processing, providing a structured way to evaluate outcomes within policy limits.

3.3.2. Probabilistic calibration (FIT-CAL-TEST)

Because policy enforcement depends on calibrated probabilities, all models were calibrated under the time-respecting FIT-CAL-TEST protocol (Guo et al., 2017). The models were trained on the FIT split, calibrated on the temporally adjacent CAL split, and evaluated on TEST, ensuring chronological integrity and avoiding calibration leakage. The calibration methods were model-specific:

- **LR:** Sigmoid (Platt) scaling, correcting mild miscalibration from regularization or class imbalance.
- **RF and GBDT:** Isotonic regression provides a flexible method for calibrating discrete probability outputs generated by tree ensemble models.

The calibration used CalibratedClassifierCV (cv = "prefit") to support post-training calibration.

All calibration metrics (ECE, ACE, MCE, and Brier score) were computed using equal-mass (quantile-based) binning with 20 bins and bootstrapped 100 times to obtain 95 % CIs. These distributions were later presented in a table and referenced by the policy gatekeeper for conservative decision-making. Reliable calibration is critical in educational AI: mis-calibrated dropout probabilities may distort institutional resource allocation, making probabilistic integrity as essential as raw predictive accuracy.

3.3.3. Fairness mitigation toolbox

To evaluate equity under policy constraints, we integrated two complementary fairness mitigation strategies, operating at different stages of the pipeline. This multi-granular toolbox reveals the trade-offs among performance, calibration, and fairness, clarifying which metrics form binding constraints and informing policy-tuning guidance. No pre-processing mitigation was applied in this study. The fairness interventions were restricted to in-processing and post-processing stages, enabling a clear separation between predictive learning and governance-driven constraint enforcement.

- **Post-processing (ThresholdOptimizer-TO, Fairlearn):** Applied to model outputs, TO optimize group-specific decision thresholds on predicted probabilities to minimize empirical classification error subject to fairness constraints. In this study, TO enforced either DP or EO as specified by the active policy tier, without retraining or modifying the underlying model parameters. TO was applied independently across the protected attribute axes gender, imd_band, highest_education, and the intersectional subgroup (gender × imd_band). This method enables institutions to add fairness compliance to existing or external models (Weerts et al., 2023).
- **In-processing (ExponentiatedGradient-EG, Fairlearn):** Applied during training for LR, EG minimizes empirical classification loss while enforcing fairness limitations through a constrained optimization formulation. In this implementation, the objective corresponds to logistic loss subject to either DP or EO constraints, with a fixed fairness slack parameter controlling the accuracy-fairness trade-off. EG produces a convex mixture of hypotheses, ensuring stable convergence and interpretable coefficients while integrating fairness directly into the training process (Agarwal et al., 2018).

Fairness gaps (DP, EO, and intersectional) were computed with bootstrap CIs and enforced only for subgroups with at least 30 students ($n \geq 30$), a commonly used threshold to avoid unstable rate estimates in subgroup fairness analysis. No smoothing, imputation, or post-hoc aggregation of subgroup rates were applied. For each configuration, the gatekeeper logged the worst-case intersectional gap and the corresponding shortfall magnitude, ensuring that compounded inequities were visible even when not binding. This integrated toolbox provides institutions with both rapid-deployment fairness corrections (TO) and deeper, principle-driven mitigation options (EG). The fairness mitigation strategies described above give rise to a total of sixteen model-policy configurations. For each of the three base classifiers (LR, RF, and GBDT), one calibrated baseline configuration without fairness mitigation was evaluated, together with four post-processing TO configurations corresponding to the protected attribute axes gender, imd_band, highest_education, and the intersectional attribute gender × imd_band. This generates fifteen configurations across the three models. In addition, one in-processing EG configuration was evaluated for logistic regression under the intersectional constraint.

Fairness mitigation was not optimized jointly across multiple protected attributes. Instead, each protected axis was evaluated independently, with the intersectional attribute (gender × imd_band) treated as a distinct constraint rather than an aggregate of marginal gaps. No weighted averaging or conflict-resolution heuristic was applied at the mitigation stage. The trade-offs across attributes were resolved exclusively at the governance level, where the gatekeeper enforced compliance using the worst-case violation across all protected and intersectional groups. No fairness mitigation was applied beyond these predefined strategies, and no subgroup-specific hyperparameter tuning was performed.

3.3.4. Evaluation signals and policy flags

For each of the sixteen predefined (model, calibration, mitigation, protected axis) configurations, the framework computed a complete set of metrics and annotated them with policy flags:

- **Performance:** AUC, macro-F1, dropout recall.
- **Calibration:** Brier score, ECE, ACE, MCE (+95 % CIs).
- **Fairness:** DP, EO, intersectional gaps (+95 % CIs).
- **Explainability/Latency:** SHAP coverage, P95 explanation latency.

Each metric was compared against thresholds defined in the PaC schema (Table 3). The configurations were annotated with Boolean flags (e.g., performance_ok, fairness_gender_ok) and logged in the audit ledger. The violations triggered diagnostic shortfall entries that specify the exact metric, its CI bounds, the threshold, and the numerical gap. This design ensures that governance decisions are deterministic, uncertainty-aware, and fully traceable, enabling both automated enforcement and transparent institutional review.

3.4. Gatekeeper enforcement

The gatekeeper operates PaC by transforming governance thresholds into deterministic, auditable decisions that determine whether a model configuration is eligible for deployment. By binding approvals directly to explicit, machine-readable policies, the gatekeeper eliminates subjective judgment and ensures that all decisions are transparent, reproducible, and cryptographically verifiable through the hash-chained governance ledger.

3.4.1. Decision semantics

The gatekeeper executes a deterministic, auditable sequence configuration for each candidate (model, calibration method, and mitigation strategy):

- **Load Artifacts:** Inputs include the evaluated metrics JSON (performance, calibration, fairness, and explainability results), the SHAP explainability log, and the existing governance audit chain to maintain provenance continuity.
- **Compare to Active Policy:** The active policy threshold covering performance, calibration, fairness, explainability, and operational latency, are loaded from the corresponding policy JSON file. Both per-axis fairness (gender, imd_band, highest_education) and intersectional fairness (gender × imd_band) are enforced, provided subgroup size ≥30 to ensure statistical validity.
- **Apply CI-aware Decision Rule:** Each metric is evaluated against its bootstrapped 95% CI, applying the conservative rule (using lower bounds for "≥" and upper bounds for "≤" metrics). A configuration is APPROVED only if all evaluation flags, performance_ok ∧ calibration_ok ∧ fairness_ok ∧ explainability_ok ∧ ops_ok, are true under this conservative mode.
- **Select Best Policy-Compliant Configuration:** When multiple configurations satisfy a policy, the current implementation identifies the best-performing compliant configuration primarily based on

maximizing AUC. The framework supports advanced tie-breaking, so future policies could rank metrics like dropout recall, ECE, or macro-F1 according to educational priorities, such as valuing learner safety over model efficiency.

- **Log reasons and artifacts:** All outcomes are recorded as machine-readable entries appended to the immutable hash-chained audit log, including policy ID, configuration hash, timestamp, metric shortfalls, and the SHA-256 digest of each dependent artifact.

This CI-driven decision process guarantees that approvals are supported by statistics, completely traceable, and remain consistent even during repeated runs.

3.4.2. Diagnostic mode

Beyond binary enforcement, the gatekeeper supports a diagnostic mode that quantifies shortfalls relative to policy thresholds for every metric. Instead of a simple "BLOCK," the system explains why a configuration failed and by how much. Each violation includes both its absolute deviation and confidence bounds, allowing developers and educators to pinpoint whether improvements should target recall, calibration, or subgroup equity. Each attempt is recorded in the governance ledger, ensuring that exploratory adaptation remains accountable and auditable. All diagnostic information feeds into three analytic tables that extend the governance framework beyond binary enforcement:

- **Metrics + 95 % CI:** Presents all key metrics with CIs across models and policies, addressing statistical uncertainty.
- **Shortfall Heatmap:** Quantifies distances from thresholds for every metric-policy pair, revealing binding constraints and key bottlenecks.
- **Policy Tuning Suggestions:** Identifies which constraints prevented approval and estimates the minimal relaxation (in %) required to achieve compliance.

These diagnostics transform the gatekeeper into an evidence-driven decision engine capable of guiding responsible adoption.

3.4.3. Structured reporting and outputs

To ensure transparency and reproducibility, the gatekeeper generates a set of structured outputs at the end of each run:

- **Best Unconstrained vs. Best Compliant Model:** Highlights trade-offs in performance, calibration, and fairness between unrestricted and policy-compliant configurations.
- **Gatekeeper Outcomes and Block Reasons:** Lists all model mitigation axis combinations with their APPROVE/BLOCK decisions and categorized failure reasons.
- **Governance Overhead Summary:** Reports P95 event latencies, decision times, and audit log sizes across repeated runs.
- **Diagnostic Extensions:** Capture CI-based metrics, shortfall distributions, and policy-tuning insights, enabling stakeholders to understand feasibility boundaries.
- **Final Governance Report:** Consolidates policy definitions, compliance outcomes, decision rationales, and artifact SHA-256 hashes into a unified JSON file linked to the audit chain.

The core properties of the advanced gatekeeper include: (i) deterministic, CI-aware enforcement; (ii) fine-grained fairness auditing with intersectional protections; (iii) an immutable governance chain; (iv) dual policy tracks (strict/idealized and deployment-realistic); and (v) diagnostic accountability with quantified violations. By combining binary enforcement with rich diagnostics and immutable auditability, the gatekeeper elevates governance from a compliance checklist to a trustworthy, evidence-driven governance ecosystem that institutions can adopt, audit, and refine over time.

3.5. Audit logging

A cornerstone of the governance pipeline is its immutable, evidence-enriched audit logging system, which ensures that every event, from preprocessing to enforcement, is recorded, integrity-verified, and cryptographically chained. This subsystem transforms responsible AI principles into verifiable practice: every model approval, rejection, or threshold decision is traceable to auditable evidence.

3.5.1. Hash-chained event logging

Each stage of the pipeline, schema generation, model training, calibration, SHAP explainability, gatekeeper enforcement, diagnostic evaluation, and governance overhead analysis, emits a structured JSON event. Every event contains metadata (timestamp, stage ID, policy ID, and artifact hashes) and the SHA-256 digest of the previous event, creating a tamper-evident hash chain. This event-level chaining produces a transparent, append ledger analogous to a lightweight blockchain but without distributed consensus overhead. It provides the same tamper-evidence guarantees while remaining computationally efficient for educational institutions that require transparency without infrastructure complexity. Any modification to an earlier record immediately invalidates all subsequent hashes, ensuring that the audit trail remains cryptographically verifiable end to end.

The audit logging system is designed to provide tamper evidence rather than tamper prevention. The primary threat model assumes the possibility of post-hoc modification or selective deletion of governance evidence by privileged insiders, administrators, or downstream operators after model evaluation or deployment decisions have been made. The hash-chained design ensures that any alteration to historical records invalidates all subsequent hashes, making such tampering immediately detectable during audit verification. The system does not attempt to defend against full host compromise, collusion among all privileged operators, or denial-of-service (DoS) attacks, which are considered out of scope for this PoC. The trust is placed in the integrity of the execution environment at runtime; cryptographic chaining enables independent verification of log integrity after execution rather than real-time enforcement.

3.5.2. Artifact integrity verification

All major governance artifacts are integrity-protected with SHA-256 hashes and linked to their corresponding audit entries. This guarantees that every dataset, policy, and diagnostic table can be independently validated by auditors. The artifacts include, preprocessing schema snapshots, consent mask files and leakage-free datasets, sensitive-attribute datasets for fairness auditing, calibration and fairness evaluation logs, SHAP explainability outputs and latency metrics, diagnostic Tables: metrics + CI, shortfall heatmap, and policy tuning, policy JSONs and threshold-source rationales and final governance report. The hash value of each artifact is recorded alongside the audit event that generated it, enabling cross-verification between metadata and content. This ensures that both internal and external auditors can reproduce outcomes with verified integrity of every intermediate result.

3.5.3. Governance overhead monitoring

To show that accountability can coexist with efficiency, the audit subsystem also monitors governance overhead across the pipeline. The following performance indicators are logged:

- **Event-level Latency:** Median and P95 processing time for each audit event.
- **End-to-end Latency:** Total wall-clock duration from preprocessing to final report generation, benchmarked against the operational thresholds defined in the policy schema.
- **Storage Footprint:** Cumulative artifact size produced during pipeline execution.

All metrics are exported in JSON (for detailed technical inspection) and CSV (for institutional summaries). This dual-format reporting allows technical auditors to assess process-level performance, while enabling non-technical governance boards to track compliance costs and feasibility in accessible formats.

3.5.4. Chain head and final governance report

Each audit sequence maintains a chain head, the SHA-256 hash of the latest event, which acts as a cryptographic fingerprint for the entire audit history. At pipeline completion, the system produces a final governance report that consists of gatekeeper decisions (approved/rejected) with CI-aware confidence bounds, machine-readable reasons for any threshold violations, pointers to all logged artifacts and their integrity proofs, summaries of governance overhead (latency and footprint) and the chain head digest, serving as a single verifiable checksum for the complete audit log.

The audit ledger is operated by the deploying institution (e.g., an institutional analytics or IT governance unit). Writing access is restricted to the governance pipeline itself, while read access may be granted to auditors, ethics committees, regulators, or accreditation bodies. The audit records are append-only and retained in accordance with institutional data-governance and regulatory policies. Because the ledger stores cryptographic hashes and structured metadata rather than raw student data, long-term retention does not materially increase privacy risk. This final report acts as the single source of truth for institutional compliance verification and can be validated independently through a built-in function that recomputes and compares hashes.

3.5.5. Contribution and significance

This audit system advances beyond previous educational AI studies by combining cryptographic traceability with evidence-level reproducibility, ensuring that governance outcomes are both trustworthy and verifiable. Specifically, it provides:

- **End-to-End Tamper Evidence:** Every model lifecycle event is chained through SHA-256 for immutable traceability.
- **Cross-Artifact Integrity Verification:** All datasets, explainability logs, fairness reports, and policy files can be independently validated.
- **Evidence-Enriched Accountability:** CI-based diagnostics are integrated directly into the governance ledger.
- **Measured Governance Feasibility:** Real-time latency and storage metrics show that accountability introduces manageable operational costs.

These capabilities establish a verifiable provenance layer for educational AI governance, turning transparency, fairness, and auditability from principles into operational guarantees that can withstand institutional and regulatory scrutiny.

3.6. Operational metrics

Beyond predictive accuracy, calibration, and fairness, the governance framework evaluates operational metrics that determine whether the system is deployable in real educational environments. The pipeline integrates explainability, efficiency and governance overhead into policy enforcement, along with performance and fairness, ensuring models remain equitable, reliable, and scalable for institutional use.

3.6.1. Explainability efficiency

Explainability was implemented using SHAP, a model-agnostic method that attributes feature contributions at both local (per student) and global (group) levels. To make interpretability auditable and enforceable, two quantitative metrics were evaluated and encoded in the policy schema:

- **Coverage:** The proportion of TEST instances for which SHAP values were successfully computed. The high coverage ensures that transparency extends to nearly all predictions, not a limited subset. The policies mandated $\geq$0.95 coverage under strict and $\geq$0.80 under lenient settings, enforcing the institutional principle that every at-risk learner should ideally have an accompanying explanation. The coverage reflects the existence of a valid local explanation for each prediction and does not, by itself, assess the semantic usefulness or human interpretability of those explanations.
- **Latency (P95):** The 95th percentile computation time for a single-row SHAP explanation, averaged over multiple repetitions to stabilize variance. Under strict policies, latency was capped at 250 ms; under lenient tiers, 600 ms. These thresholds intend to guarantee that even worse-case explanations remain interactive and suitable for real-time dashboards used by advisors. SHAP latency was measured as the wall-clock time required to compute a local explanation for a single prediction instance, excluding one-time initialization costs such as model loading and explainer setup. The latency statistics are reported as percentiles (P50, P95) computed across all explanation events generated during TEST set evaluation. The reported P95 value therefore reflects worst-case per-instance explainability latency under normal pipeline execution.

In this PoC, explainability is treated as a governance and auditability requirement rather than a measure of human decision support quality. Accordingly, the framework verifies that every prediction is accompanied by a theoretically faithful explanation, and that such explanations can be generated within operational latency bounds. We do not claim that high coverage implies explanation usefulness, stability, or ease of interpretation for end users. Metrics such as perturbation-based fidelity, explanation stability under input noise, or human interpretability assessments fall outside the scope of this PoC. Both metrics were logged into the governance audit ledger and their compliance status factored into the gatekeeper decisions. The shortfalls related to policy thresholds for these metrics were included in Shortfall Heatmap. This elevates explainability from an optional interpretability tool to a governance-enforced operational standard. The explanation fidelity was not evaluated using perturbation-based or surrogate-model metrics in this PoC. Instead, fidelity is implicitly guaranteed using exact SHAP formulations for linear and tree-based models, which are theoretically faithful to the underlying prediction function. The formal empirical or human-centered fidelity evaluations are left for future work.

3.6.2. Governance overhead

To ensure that accountability remains feasible, the framework continuously monitors the computational cost of governance itself. Three policy-bounded indicators were recorded:

- **Event Latency (Median, P95):** Wall-clock time required to record a single audit event and update the cryptographic hash chain. The percentiles are computed across all audit events emitted during a pipeline execution.
- **Pipeline Wall-Time:** End-to-end runtime from preprocessing through final gatekeeper decision, measured once per pipeline run.
- **Artifact Footprint:** Total on-disk size of logs, datasets, and reports produced by the governance pipeline.

These values were benchmarked against the operational thresholds defined in the PaC schema and evaluated under the ops_ok flag in the gatekeeper. If the governance subsystem exceeded policy limits (e.g., P95 $>$ 350 ms), the configuration was automatically blocked even if its predictive and fairness metrics passed. In this study, the reported governance latency value (P95 $\approx$ 23.5 ms) corresponds to the 95th percentile event-level audit-logging latency, indicating that 95 % of governance events completed within 23.5 ms. All overhead metrics were exported in both hierarchical JSON (for technical auditors) and flattened CSV (for institutional dashboards) to balance precision and accessibility.

3.6.3. Integrated enforcement

Explainability and governance overhead metrics were fully integrated into gatekeeping enforcement. The configurations failing SHAP coverage, latency, or overhead requirements were automatically rejected, and violations were recorded as shortfall entries in the audit log and aggregated in tables. This integration represents a methodological advance for educational AI:

- Explainability coverage and latency are first-class governance constraints, not after-the-fact diagnostics.
- SHAP latency estimates are stabilized through repeated runs with CI-based reporting.
- Governance overhead is explicitly quantified and policy-bounded, highlighting that fairness, calibration, and auditability can coexist without prohibitive cost.

By embedding these operational criteria directly into policy enforcement, the framework addresses the gap integrating responsible-AI principles and real-world deployment feasibility, an aspect largely absents from previous dropout-prediction research.

3.7. End-to-end pipeline algorithm

To consolidate the methodological design, Algorithm 1 presents the complete governance-aware dropout-prediction pipeline. It integrates every component, policy schema initialization, preprocessing, temporal splitting, calibration, fairness mitigation, Explainability, gatekeeper enforcement, and audit logging, into a single deterministic, CI-aware, and auditable workflow.

Every approval or rejection is traceable to explicit thresholds, logged artifacts, and hash-chained evidence.

Algorithm 1. Policy-Governed Pipeline for Dropout Prediction

```
PROCEDURE PIPELINE (T, P, S, F_exclude, C, M, G):
1. AUDIT_INIT:
  - Summarize raw tables (columns, dtypes, counts).
  - Record schema snapshot in audit log (hash-chained).
2. PREPROCESS:
  - Merge OULAD tables into a student-level data set.
  - Construct target y (1 = Withdrawn/Fail; 0 = Pass/Distinction).
  - Retain sensitive attributes S for fairness auditing.
  - Exclude leakage fields (final_result, date_unregistration).
  - Standardize categoricals; encode missing values (0/sentinels).
  - Save features-only and full datasets, record artifact hashes.
3. TIME-RESPECTING SPLITS:
  - For each module: latest → TEST; preceding → CAL; earlier → FIT.
  - Record split indices in audit log.
4. TRAIN + CALIBRATE (per model m ∈ M):
  - Train base model on FIT.
  - Calibrate on CAL using Platt (LR) or isotonic (RF/GBDT).
  - Save calibrated model; log metadata.
5. MITIGATE + EVALUATE (per mitigation g ∈ G; per axis a ∈ S ∪ {None}):
  - Apply fairness mitigation (TO or EG).
  - Predict on TEST; compute metrics:
    • Performance: AUC, macro-F1, dropout recall
    • Calibration: Brier, ECE/ACE/MCE (+95 % CI)
    • Fairness: DP/EO gaps (+95 % CI); worst intersectional gap
    • Explainability: SHAP coverage, P95 latency
    • Governance overhead: event latency, wall-time, artifact size
  - Assign policy flags (e.g., performance_ok, fairness_gender_ok).
  - Record results → 'results_per_config.json'.
6. GATEKEEPER:
  - Approve if all flags are true under policy P (CI-aware), else block.
  - Log violation reasons; append to audit chain.
7. REPORTING:
  - Generate Tables (compliance, metrics + CI, shortfalls).
  - Output final 'governance_report_Advanced.json' with artifact hashes.
8. RETURN:
  - Approved set A; final chain head H*; complete governance ledger.
```

Unlike traditional educational AI pipelines focused solely on accuracy or fairness auditing, this algorithm treats governance as a computational contract. Every stage is logged and hash-chained for tamper-evidence, evaluated under time-respecting splits that mimic deployment, and constrained by policy-encoded thresholds covering performance, calibration, fairness, explainability, and operational overhead. By uniting statistical validity with institutional accountability, the pipeline provides a deployment-ready blueprint for responsible AI in education, one where every decision is explainable, auditable, and trustworthy.

4. Results

The results are organized into the following subsections reflecting the sequential logic of the governance-audited pipeline. First, the OULAD dataset is reconstructed as a governed artifact under explicit consent and leakage-control policies. Next, unconstrained baseline performance is established before progressively integrating fairness mitigation, gatekeeper compliance evaluation, and operational feasibility tests. Each subsection presents quantitative metrics, compliance margins, and diagnostic artifacts, showing how the pipeline transforms predictive modeling into verifiable governance evidence.

4.1. Dataset as a governed artifact

To ensure that all subsequent analyses were reproducible and aligned with the governance framework, the OULAD dataset was processed under the consent-filtering and leakage-control procedures. As presented in Table 4, a simulated consent mechanism was introduced due to the absence of an explicit consent field in the original OULAD data. Using a fixed random seed (42) to ensure reproducibility, this simulation excluded 3227 learners, approximately 10 % of the group. The final sample of 29,366 learners served as the consistent basis for all modeling, fairness auditing, and explainability assessments. This simulated consent mechanism is used solely to show how consent limitations

Table 4
Consent simulation and leakage-control summary for the dataset (post-consent snapshot).

Item	Value	Notes
Consent required	Yes	Governance setting enforced in this run
Source consent column present	No	Consent simulated (original OULAD lacks explicit consent field)
Consent simulation seed	42	Ensures reproducibility
Rows removed by consent simulation	3227	≈10% removed before modeling/auditing
Rows kept (post-consent)	29,366	Used in all results and audits
Proportion kept	90.1%	As logged by audit pipeline
Pre-consent size (implied)	32,593	Sum of rows kept + removed
Sensitive axes audited	gender; imd_band; highest_education	Fixed across all runs
Excluded leakage fields	final_result; date_unregistration; target; student_consented	Removed before modeling; last two are governance-added
Features artifact	OULAD_cleaned.csv	Post-consent, leakage-safe features
Full artifact (with sensitive)	OULAD_cleaned_full.csv	Post-consent, retains audited axes
Features shape (rows × cols)	29,366 × 15	Modeling input
Full shape (rows × cols)	29,366 × 19	Includes sensitive + governance fields

can be operationalized within the governance pipeline and does not represent actual consent decisions by learners.

The sensitive axes, gender, imd_band, and highest_education, were fixed as governance-relevant attributes for this study, enabling consistent subgroup-level fairness auditing across all experiments. In parallel, potential leakage sources such as final_result and date_unregistration were excluded before model fitting. This dual artifact ensured transparency: the first artifact contained only leakage-safe features for modeling, while the second preserved sensitive attributes for auditing and interpretability verification. The resulting dataset (29,366 × 15 for the feature set; 29,366 × 19 for the full artifact) serves as an auditable checkpoint for institutional reproducibility. By integrating consent logic, exclusion traceability, and sensitive-attribute retention directly within the preprocessing pipeline, the framework operationalizes the concept of data as a governed artifact, a foundation for all downstream policy enforcement and explainability evaluation. The governed dataset provides a verified consent history and secure feature schema for the analyses in Section 4.2.

4.2. Baseline model performance

To establish a reference point before applying governance constraints, three baseline classifiers, LR, RF, and GBDT, were trained and evaluated on the time-respecting FIT-CAL-TEST split. The evaluation covered predictive performance, calibration reliability, and fairness parity across sensitive axes (gender, imd_band, and highest_education). Table 5 reports point estimates with 95% bootstrap CIs (100 resamples) under the FIT-CAL-TEST split. The bootstrapping is used to quantify uncertainty; governance thresholds are evaluated against point estimates in standard mode and against CI bounds in conservative mode. The metrics include predictive utility (AUC, macro-F1, recall for dropout), calibration (ECE and Brier score), and subgroup fairness (DP and EO gaps). These baselines represent the unconstrained upper bounds of model accuracy before any fairness mitigation or policy enforcement.

All three models showed strong predictive utility, indicating that the OULAD dataset supports effective predictive modeling under the evaluated feature set and split. The LR baseline achieved the highest discrimination (AUC = 0.920 [0.915-0.924]) and stable calibration (ECE = 0.035 ≤ 0.05), satisfying the calibration requirement of the strict policy for this experimental setting. The comparisons among baseline models were interpreted using bootstrap CIs rather than parametric significance tests; overlapping intervals indicate comparable discrimination, while non-overlapping intervals indicate statistically meaningful differences under uncertainty-aware evaluation. The GBDT model achieved the best dropout recall (0.781 [0.770-0.793]) but showed the poorest calibration (ECE = 0.066 > 0.05). Despite high accuracy, fairness audits exposed consistent violations of the evaluated institutional policy thresholds across multiple sensitive axes. Under the strict policy, all models failed across multiple dimensions:

- **Highest Education**: EO gaps up to 0.174 (RF) > 3 × threshold.
- **IMD Band**: EO gaps ≈0.146 (LR) and 0.144 (GBDT), indicating socioeconomic sensitivity bias.
- **Gender**: Minor but non-negligible gaps ≈0.09-0.10.

These results show that, in this experimental setting, models with strong predictive accuracy can still fail to meet fairness standards, and that accuracy alone is insufficient as a deployment criterion. Fig. 4 shows the fairness-performance frontier for the unconstrained baselines under the medium_realistic policy thresholds. Each point represents the trade-off of a model between dropout recall and fairness gap. Even under relaxed criteria, all three evaluated baseline models remain non-compliant, their EO gaps for imd_band and highest education remain above the permissible region. This visualization highlights that relaxing thresholds alone does not close equity gaps. Instead, it confirms the necessity for active fairness mitigation, which will be examined in Section 4.3.

4.3. Impact of fairness mitigation

Having established the performance and fairness shortfalls of the unconstrained baselines, we next examined how fairness mitigation affects compliance trajectories. All mitigation strategies were applied to models that had already undergone post-hoc probabilistic calibration under the FIT-CAL-TEST protocol; no additional recalibration was performed after fairness mitigation. Two complementary strategies were evaluated:

- **ThresholdOptimizer-TO**: A post-processing approach that adjusts decision thresholds to satisfy fairness constraints.
- **ExponentiatedGradient-EG**: An in-processing reduction method that jointly optimizes accuracy and fairness during training.

Because TO can target multiple fairness axes, the results reported here correspond to its most comprehensive configuration, intersectional gender × IMD Band mitigation, which is intended to approximate realistic institutional audit scenarios within this study. The next section builds on these results by discussing complete governance enforcement through binary choices of the gatekeeper to either APPROVE or BLOCK.

The results in Table 6 show important performance-fairness trade-offs within the evaluated models and policy settings:

- **ThresholdOptimizer:** Substantially improved fairness for its target attributes, reducing, for example, the gender EO gap of LR model from 0.095 to 0.007 and IMD gap from 0.146 to 0.089, but at the expense of predictive utility. The dropout recall declined from 0.757 to 0.586 (−17.1 percentage points) and fell below the recall requirement under the Strict, Medium, and Lenient tiers; it also fails the 0.70/0.65 recall floors under the Realistic tiers. By comparison, macro-F1 remains above the 0.75 floor under the Realistic tiers, showing how fairness mitigation can satisfy balanced-accuracy targets while violating recall requirements for at-risk learners.
- **ExponentiatedGradient:** Achieved broader reductions across the audited axes in this experimental setting but introduced severe calibration and reliability penalties. For the LR model, AUC dropped from 0.920 to 0.777, and ECE rose from 0.035 to 0.226, failing both predictive and calibration standards. The non-overlapping AUC ranges between the unconstrained and mitigated setups show that the decline in predictive performance is statistically significant when comparisons account for uncertainty.

Table 5
Unconstrained performance, calibration, and fairness gaps.

Model	AUC [95 % CI]	F1 Macro [95 % CI]	Recall (Dropout) [95 % CI]	ECE [95 % CI]	Brier [95 % CI]	EO Gap (Gender)	EO Gap (IMD)	EO Gap (Highest Edu)	Worst Intersectional Gap [95 % CI]
LR	0.920 [0.915 - 0.924]	0.867 [0.861 - 0.873]	0.757 [0.748 - 0.767]	0.035 [0.031 - 0.041]	0.100 [0.096 - 0.103]	0.095	0.146	0.155	0.296 [0.257 - 0.379]
RF	0.902 [0.894 - 0.910]	0.851 [0.843 - 0.861]	0.774 [0.762 - 0.785]	0.035 [0.032 - 0.039]	0.107 [0.101 - 0.113]	0.088	0.138	0.174	0.273 [0.225 - 0.335]
GBDT	0.889 [0.880 - 0.899]	0.821 [0.812 - 0.834]	0.781 [0.770 - 0.793]	0.066 [0.060 - 0.073]	0.126 [0.120 - 0.133]	0.097	0.144	0.141	0.254 [0.210 - 0.301]

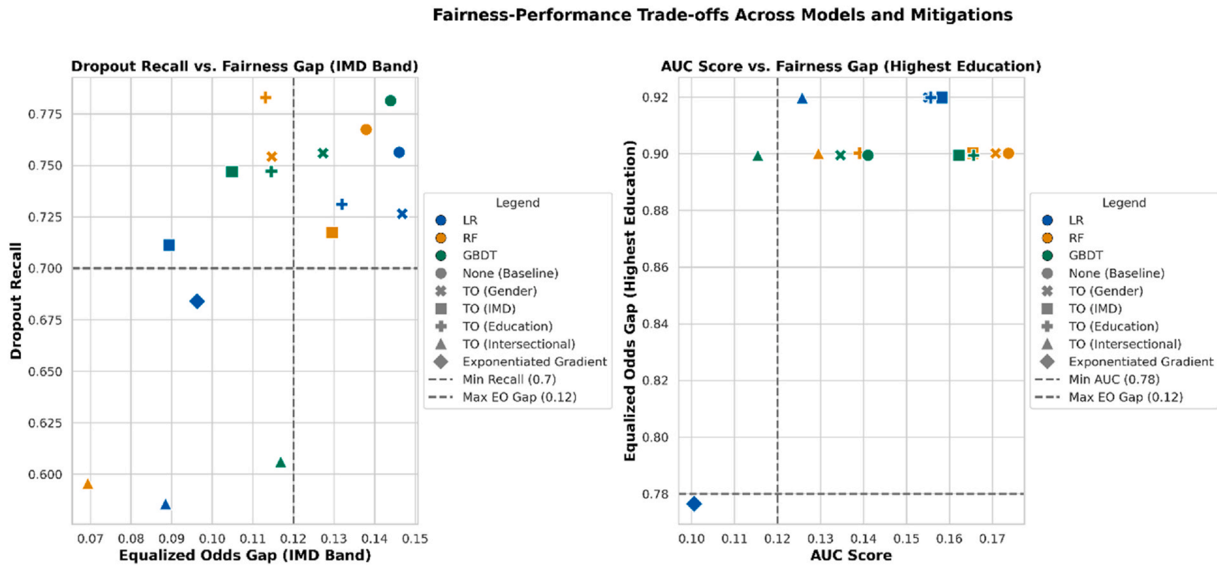

Fig. 4. Fairness-performance trade-offs across models under the medium realistic policy.

Table 6
Mitigation impact compared to unconstrained baselines.

Model	Mitigation	AUC	Macro-F1	Recall (Dropout)	ECE	Brier	EO Gap (Gender)	EO Gap (IMD)	EO Gap (Highest Edu)	Worst Intersectional Gap
LR	None	0.920	0.867	0.757	0.035	0.100	0.095	0.146	0.155	0.296
	TO (Intersectional)	0.920	0.782	0.586	0.035	0.100	0.007	0.089	0.126	0.235
	EG (Reduction)	0.777	0.774	0.684	0.226	0.226	0.056	0.096	0.101	0.219
RF	None	0.902	0.851	0.774	0.035	0.107	0.088	0.138	0.174	0.273
	TO (Intersectional)	0.901	0.774	0.602	0.035	0.107	0.024	0.069	0.129	0.282
GBDT	None	0.889	0.821	0.781	0.066	0.126	0.097	0.144	0.141	0.254
	TO (Intersectional)	0.889	0.764	0.604	0.066	0.126	0.022	0.117	0.115	0.286

- Similar patterns were observed for RF and GBDT under the evaluated mitigation configurations, where post-processing mitigations corrected specific biases but simultaneously reduced recall by ≈ 0.17-0.18, again failing the institutional performance floor.

These results highlight a key governance insight for this study: under the evaluated dataset, policy thresholds, and mitigation strategies, no single mitigation achieved simultaneous compliance across all policy dimensions. In this setting, improvements in fairness were accompanied by shifts in other metrics beyond acceptable limits, revealing the complexity of multi-objective responsible-AI optimization. The standard post-hoc calibration methods such as Platt scaling and isotonic regression were used before fairness mitigation, but not afterward. The post-hoc recalibration following TO or EG would alter decision scores or probability distributions in ways that invalidate the enforced fairness limitations, requiring mitigation to be re-run. In effect, repeated calibration-mitigation cycles would create an oscillating optimization loop rather than a stable, auditable governance outcome. The observed calibration regressions therefore reflect a structural tension between fairness-constrained decision adjustment and probabilistic fidelity, rather than an omission in the modeling pipeline. This trade-off is surfaced explicitly by the framework and handled conservatively by the CI-aware gatekeeper.

Fig. 5 shows the distance to policy compliance for major fairness metrics under the lenient_realistic policy. Each bar represents the gap between observed values and policy thresholds, with green segments denoting compliance and red segments indicating residual violations.

The chart shows that while mitigated configurations (e.g., LR + TO or RF + TO) enter near-compliant zones for gender and IMD, they remain marginally above limits for highest education and intersectional parity. Rather than constructing a continuous fairness-utility Pareto frontier, this study evaluates a discrete set of policy-relevant mitigation configurations under executable governance constraints. The trade-offs are therefore assessed through comparative tables, CI-aware diagnostics, and threshold-based visualizations, reflecting institutional deployment settings in which models must satisfy predefined policy tiers rather than optimize unconstrained utility-fairness curves. Also, CIs reveal that even minor statistical uncertainty can reverse compliance status, supporting the need for CI-aware conservative evaluation.

4.4. Policy compliance under governance

The final evaluation stage applied the governance gatekeeper, which enforces all performance, calibration, explainability, and fairness thresholds simultaneously to determine whether any configuration qualifies for deployment approval. Unlike the diagnostic views in Sections 4.2 and 4.3, this stage executes a binary policy decision, APPROVE or BLOCK, based on conservative, CI-aware comparisons. These evaluations were performed over the fixed set of sixteen predefined model-policy configurations as presented in Table 7, without any additional tuning, resampling, or configuration expansion at the policy stage.

Within the evaluated dataset, policy tiers, and model, mitigation configurations, no configuration achieved full compliance under any policy tier. All sixteen predefined model-policy configurations (LR, RF, GBDT combined with baseline calibration, TO mitigation across four protected axes, and one EG configuration for LR) were blocked,

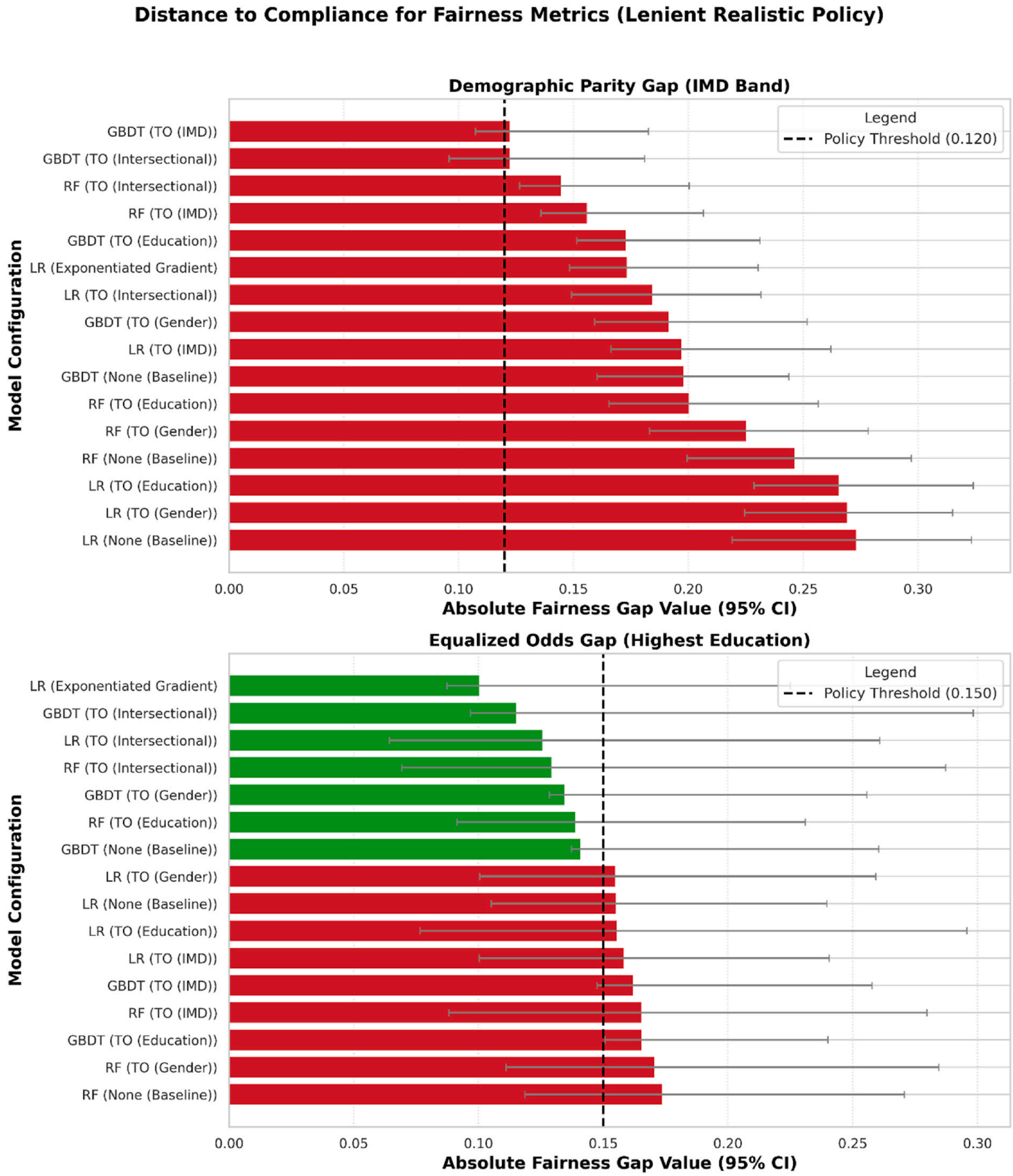

Fig. 5. Distance to Policy Compliance for Fairness Gaps under the Lenient-Realistic Policy *(Green = compliant; red = violating metrics.)*.

Table 7
Gatekeeper outcomes and top block reasons.

Policy	Configurations Tested	Configurations Approved	Configurations Blocked	Dominant Block Reasons
Strict	16	0	16	Fairness gaps on *highest_education* and *imd_band*; intersectional gap; recall
Medium	16	0	16	Fairness gaps on *highest_education* and *imd_band*; intersectional gap
Lenient	16	0	16	Fairness gaps on *highest_education* and *imd_band*; intersectional gap
Medium_realistic	16	0	16	Fairness gaps on *highest_education* and *imd_band*; intersectional gap
Lenient_realistic	16	0	16	Fairness gaps on *highest_education* and *imd_band*; intersectional gap

indicating that, under the evaluated dataset, policy thresholds, and mitigation strategies, substantial fairness mitigation was insufficient to produce a fully governable configuration once all policy criteria were enforced simultaneously. The gatekeeper logs reveal consistent, interpretable rejection patterns rather than random failures. Across every policy tier:

- Fairness violations dominate, appearing in almost every block decision.
  - The most frequent breaches involve the socio-economic attributes imd_band and highest_education, followed by the intersectional gap.
  - Even under the lenient_realistic tier, the observed biases remained far above thresholds.
- Performance and calibration metrics were also significant binding constraints. The mitigation techniques introduced severe trade-offs. For example, 6 of the 16 configurations (37.5 %) failed the strict ECE threshold, while 4 of the 16 models (25 %) failed even the Recall requirement of medium policy.
- Relaxing fairness thresholds increased tolerance but did not alter verdicts of the gatekeeper; the magnitude of inequity exceeded even the widest caps.

In this experiment, intended governance actions of the gatekeeper stop the use of highly accurate but unfair models, revealing the shortcomings of current mitigation techniques in ensuring institutions comply with fairness standards.

Fig. 6 shows the distribution of gatekeeper blocks across policy tiers. Each stacked bar represents the proportion of violations attributed to fairness, performance, calibration, explainability, and operational overhead. The plot confirms that fairness-related violations, particularly those tied to highest_education, imd_band, and intersectional subgroups, dominate across all tiers, while calibration and overhead violations remain negligible. As policies transition from strict to lenient_realistic, total block counts decrease modestly, but fairness still constitutes most rejections.

4.5. Operational governance & explainability

The final evaluation stage assessed the operational feasibility and explainability compliance of the governance pipeline itself. This analysis determines whether the framework can execute its auditing, explanation, and decision functions efficiently within institutional latency and transparency constraints. The pipeline recorded all timing and overhead metrics through the governance audit chain. As shown in Fig. 7, all evaluated operational indicators were within the limits defined by the strict policy.

- **Governance Latency:** The 95th percentile execution time across all pipeline events was 23.5 ms, an order of magnitude faster than the 150 ms ceiling.
- **Audit-chain Overhead:** The governance and auditing logic, including hash-chaining and logging, was confirmed to be highly efficient and well within policy limits.

These results validate that governance enforcement logic of the framework, including policy hashing, integrity verification, and CI-aware decision aggregation, can operate in real time without impairing scalability or user responsiveness.

The SHAP explainability audit achieved high coverage and low computational overhead.

Under the strict policy requirements (coverage $\geq$95 %, latency $\leq$250 ms), the framework attained:

- 100 % SHAP coverage for the evaluated test instances (i.e., all evaluated instances successfully explained)
- 95th percentile SHAP latency = 0.012 ms, far below the operational ceiling.

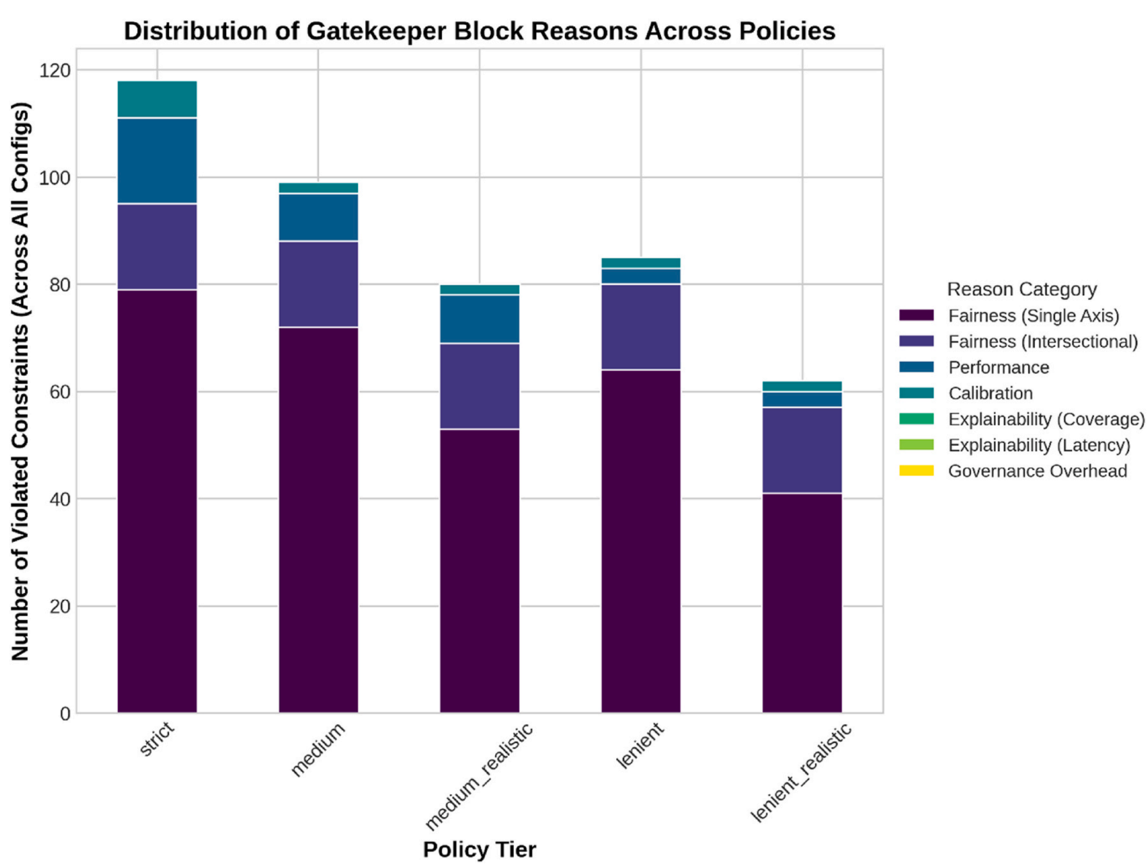

Fig. 6. Distribution of gatekeeper block reasons across policies.

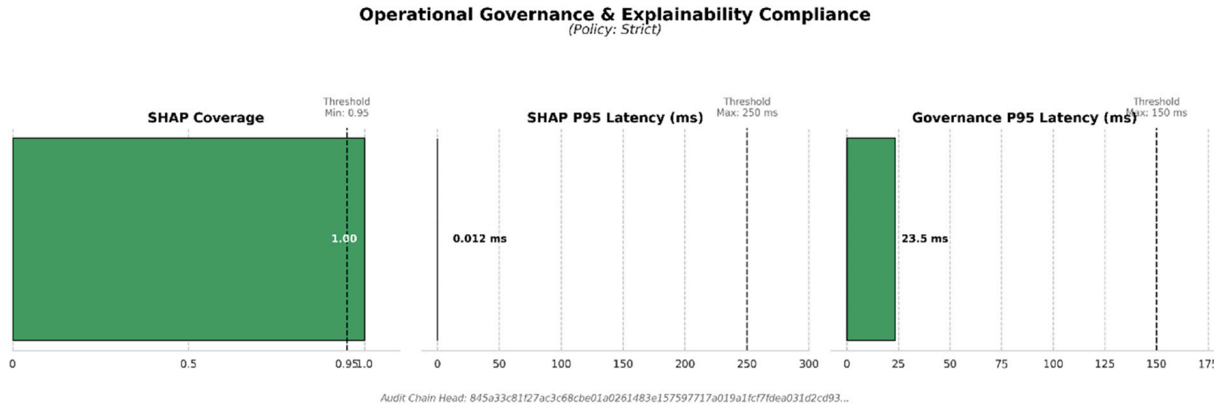

Fig. 7. Operational Governance and Explainability Compliance (Strict Policy)
(Panels show governance latency, SHAP coverage, and SHAP latency with 95 % CIs; thresholds indicated by dashed lines.).

These results indicate that, in this experimental setting, explainability is not a computational or procedural bottleneck. Moreover, the automatically generated logs provide traceable evidence of compliance, including timestamped coverage and latency distributions recorded for audit reviews. Beyond efficiency, SHAP explanations highlight the internal decision logic of the LR model, the baseline reference for explainability auditing. Table 8 lists the ten most influential features ranked by their mean absolute SHAP values.

The results highlight the behavioral dominance of temporal engagement features:the variable max_day (last interaction day) exerts more than ten times the explanatory influence of the next-ranked variable (total_clicks). However, several sensitive attributes, gender and highest_education, also appear among the top contributors, confirming their measurable role in the predictive logic of model. These findings indicate that within the evaluated governance pipeline:

- The governance pipeline achieves real-time performance with negligible computational overhead.
- The SHAP-based explainability component meets both coverage and latency requirements across all policies.
- Sensitive attributes influencing model predictions are explicitly traceable, enabling informed fairness remediation.

As a result, within this study, the dominant challenges observed are substantive (fairness alignment) rather than operational (latency or overhead). This confirms that the governance enforcement mechanisms are technically feasible and transparent, directing the way for diagnostic refinement in Sections 4.6-4.7.

Table 8
Top 10 most influential features from SHAP analysis.

Rank	Feature	Source Attribute	Mean |SHAP| Value
1	max_day	Last Interaction Day	4.629
2	total_clicks	Total VLE Clicks	0.384
3	gender_M	Gender	0.253
4	num_vle_activities	VLE Activities	0.222
5	highest_education_Lower Than a Level	Highest Education	0.204
6	studied_credits	Studied Credits	0.189
7	age_band_0-35	Age	0.154
8	gender_F	Gender	0.133
9	disability_Y	Disability	0.111
10	min_day	First Interaction Day	0.098

4.6. Diagnostic governance insights: shortfalls and tuning paths

While Section 4.4 showed that all evaluated configurations were blocked under the tested policy tiers, the diagnostic extensions in the pipeline (Tables 9 and 10) provide transparent, quantitative insight into why each model failed and how policy thresholds could be responsibly adapted. These diagnostics transform the gatekeeper, within this framework, from a binary evaluator into an explainable governance engine. Table 9 presents a shortfall heatmap for representative models, showing the precise numerical distance from policy compliance. The positive values indicate a violation, while negative values indicate the model who passed that specific test.

Table 9 reveals that the primary blockers are not uniform. While the unmitigated model fails broadly on fairness, the mitigated models expose the direct trade-offs:

- LR (TO - Intersectional) sees its Recall become the primary blocker.
- LR (EG) successfully passed the Recall test but now fails on ECE and Intersectional Gap.
- All mitigated models evaluated here (TO and EG) fail to resolve the Intersectional Gap violation, which remains the most persistent blocker.

Table 10 translates these shortfalls into actionable tuning suggestions, showing the minimum relaxation required for the single biggest blocker (the binding constraint) to be approved.

This table highlights that the main compliance issue in this study is the intersectional fairness constraint rather than marginal performance or calibration deficiencies. Achieving compliance under the Medium (Realistic) tier would require substantial relaxations of the intersectional gap threshold, ranging from +82.7% to +138.5% across the evaluated models. The performance and calibration constraints, while non-negligible, require comparatively smaller but still meaningful adjustments. For example, the LR (TO) configuration would require a 16.3% reduction in the dropout-recall threshold, while the LR (EG) configuration, although primarily blocked by intersectional fairness, would additionally require an 88.3% relaxation of the ECE limit under the Medium (Realistic) tier (from 0.12 to 0.226).

4.7. Governance traceability and Audit Integrity

The final layer of evaluation examined the traceability, integrity, and reproducibility of the governance process itself. While previous subsections evaluated model-level compliance, this stage verifies that every decision and artifact, data preprocessing, model training, explainability, fairness auditing, and policy enforcement, is cryptographically auditable and statistically defensible.

Table 9
Shortfall heatmap: Distance from policy thresholds.

Model	Policy	EO Gap (Gender) Δ	EO Gap (IMD) Δ	EO Gap (Highest Edu) Δ	Intersectional Gap Δ	ECE Δ	Recall Δ	Dominant Binding Constraint
LR (Unmitigated)	Strict	+0.045	+0.096	+0.105	+0.236	−0.015	−0.006	Fairness (Highest Edu)
LR (TO - Intersectional)	Medium Realistic	−0.113	−0.031	+0.006	+0.115	−0.085	+0.114	Recall
LR (EG)	Lenient Realistic	−0.094	−0.053	−0.049	+0.069	+0.046	−0.034	Intersectional Fairness
RF (TO - Intersectional)	Lenient Realistic	−0.026	−0.001	+0.009	+0.132	−0.145	+0.053	Intersectional Fairness
GBDT (TO - Intersectional)	Lenient Realistic	−0.028	−0.003	−0.005	+0.136	−0.114	+0.043	Intersectional Fairness

Table 10
Policy-tuning suggestions: Minimum relaxation required for compliance.

Model	Policy	Metric to Relax	Current Value	Policy Limit	Required Relaxation (%)	New Threshold for Approval
LR (TO - Intersectional)	Medium Realistic	Recall (Dropout)	0.586	0.70	−16.3%	0.586
LR (EG)	Medium Realistic	Intersectional Gap	0.219	0.12	+82.7%	0.219
RF (TO - Intersectional)	Medium Realistic	Intersectional Gap	0.282	0.12	+135.3%	0.282
GBDT (TO - Intersectional)	Medium Realistic	Intersectional Gap	0.286	0.12	+138.5%	0.286

4.7.1. Hash-chained governance audit trail

Each execution of the pipeline generated a new event entry in the immutable governance audit log, automatically linked to the previous record through SHA-256 hash. This mechanism ensures that every action, from data preparation to final gatekeeper verdict, is recorded in an append-only chain, preventing retroactive modification or deletion. The key attributes of the audit chain include:

- **Cryptographic linkage:** Each event contains prev_hash, policy_hash, timestamp, and event type (e.g., Preprocessing, Explainability, Gatekeeper). This produces a tamper-evident lineage of governance actions.
- **Dual output formats:** Both structured JSON and tabular CSV are generated, providing machine-readable and human-readable audit views for institutional recordkeeping.

An excerpt from the final audit head confirms complete policy lineage, including SHA-256 digests for every threshold configuration and model decision, ensuring cryptographic continuity between evaluation runs.

4.7.2. Uncertainty-aware governance evaluation

To further enhance reliability, the gatekeeper now operates in CI-aware mode, using the bootstrapped 95 % CIs computed in Table 9 to make statistically conservative decisions:

- For metrics where higher is better (e.g., AUC, Recall), the lower bound of the CI is compared to the policy threshold.
- For metrics where lower is better (e.g., EO Gap, ECE), the upper bound is used.

This design ensures that no configuration is approved unless compliance is statistically robust, mitigating risks of false positives due to sampling variability. All CI-aware evaluations and resulting flags are logged in the audit chain with their associated confidence levels and decisions (decision_conservative = True/False). These results indicate that, under the evaluated conditions, the governance mechanism operates in a verifiably trustworthy manner. All evaluation events are cryptographically chained, every policy configuration is hash-anchored, and decisions are made using uncertainty-aware criteria. This combination enables institutional reviewers or auditors to independently reproduce, verify, and trace evaluated model decisions without ambiguity or manual intervention.

Sections 4.6-4.7 show that the framework not only audits models but also audits itself, closing the governance loop within the proposed framework through traceability, accountability, and reproducibility. This capability advances the state of responsible AI in education by showing that trustworthiness can be operationalized, measured, and preserved across the full lifecycle of AI deployment.

The results show, in this study, the end-to-end capability of the proposed governance-audited Trust-AI framework to transform conventional educational predictive modeling into a fully traceable, policy-aware system. The governed OULAD dataset established reproducible consent and leakage safeguards (Section 4.1), while baseline analyses confirmed that even highly accurate models violated fairness criteria (Section 4.2). The subsequent mitigation experiments (Section 4.3) and gatekeeper enforcement (Section 4.4) revealed the complex trade-offs and decisive challenges to compliance. While equity gaps were the most persistent challenge, mitigation techniques introduced severe performance (Recall) and calibration (ECE) penalties that also served as primary blockers. The operational testing confirmed that the pipeline executes explainability and audit routines efficiently within strict latency budgets (Section 4.5). New diagnostic modules (Sections 4.6–4.7) quantified the precise distances to compliance, suggested responsible policy relaxations, and cryptographically linked every event to ensure governance integrity. These findings indicate that, within the evaluated dataset, policies, and modeling choices, the primary challenges observed in building trustworthy educational AI systems are substantive and multi-dimensional, achieving fairness, performance, and calibration simultaneously, rather than technical (scalability or audit cost), setting the stage for the following Discussion section, which interprets the institutional and ethical implications of these results.

5. Discussion

The findings presented in this study show that executable, PaC governance for educational AI is feasible and, in this context, critically important for closing the gap between abstract ethical principles and concrete institutional practice. This study moves beyond theoretical post-hoc analysis to an operational model of ex-ante enforcement. However, the central contribution is not just the creation of a “gatekeeper” that blocks non-compliant models. The central novelty lies in highlighting a diagnostic governance engine within this study. We showed that the real challenge is not simply building a model that passes a threshold, but navigating the severe, multi-dimensional trade-offs that emerge. The productivity of our framework lies in its ability to transform a “BLOCK” verdict from a technical dead end into an actionable, evidence-based institutional conversation. It provides a practical blueprint for creating AI systems that are accountable, transparent, and, above all, justifiable. Box 1 shows a sample governance audit report that clearly communicates gatekeeper decisions, key metrics, and suggested actions for institutional review to non-technical stakeholders.

5.1. Interpreting the findings: answering the research questions

This study was guided by five research questions. We interpret the results below, emphasizing what is new relative to previous dropout-prediction research: ex-ante, PoC enforcement; quantified diagnostic gatekeeping; and auditable, low-overhead governance.

5.1.1. RQ1: how can institutional AI governance policies be encoded as executable rules that support consistent, auditable enforcement?

Our results indicate that treating governance as PaC enables deterministic and auditable enforcement in the evaluated framework. The institutional requirements were encoded in versioned JSON policies that specify thresholds for performance (e.g., AUC), calibration (e.g., ECE), fairness (e.g., EO Gaps), and operations (e.g., SHAP latency). These policies served as the single source of truth for the automated gatekeeper. Every decision and its corresponding evidence were written to a hash-chained audit log, generating compliance-grade traceability. This ex-ante mechanism binds deployment to computable rules, ensuring every model is held to the same auditable standard.

5.1.2. RQ2: under a time-respecting evaluation, what trade-offs arise among performance, calibration, and fairness, and how should institutions reason about them?

Our results show that trade-offs are not only present but are severe, complex, and multi-directional in the evaluated models and policy settings. Different mitigation strategies introduce entirely different “failure signatures.” For example, in-processing mitigation (EG) caused a substantial calibration failure (ECE rising from 0.035 to 0.226) and a significant AUC drop. All reported trade-offs were interpreted under uncertainty-aware comparison rather than formal hypothesis testing. The changes in predictive utility and calibration (e.g., AUC and ECE) were assessed using bootstrap CIs; non-overlapping intervals between unconstrained and mitigated configurations indicate statistically meaningful degradation, while overlapping intervals indicate comparable performance under uncertainty. Rather than optimizing a continuous fairness-utility Pareto frontier, the framework evaluates a discrete, policy-constrained configuration space, reflecting institutional deployment settings in which models must satisfy predefined governance thresholds rather than maximize unconstrained utility. In comparison, post-processing (TO) preserved calibration but decimated model utility (Dropout Recall falling from 0.757 to 0.586). The contribution of our framework is not to solve this tension but to make these distinct trade-off profiles visible and quantifiable. The gatekeeper allows leaders to reason about concrete, evidence-based choices (e.g., *“Which is a worse outcome: a model that is poorly calibrated or a model that fails to identify 17% more at-risk students?”)*. By making the “cost” of fairness explicit, the framework enables principled, evidence-based decisions.

5.1.3. RQ3: does a multi-strategy mitigation toolbox (post-processing and in-processing) improve feasibility under realistic policy thresholds compared with a single-method approach?

In this study, a multi-strategic toolbox was insufficient to achieve compliance with any model configuration. While mitigation strategies did reduce specific fairness gaps, they were not sufficient to achieve full compliance. As noted in RQ2, these techniques introduced severe, negative trade-offs that resulted in the models being blocked for violating other policy thresholds (e.g., performance or calibration). Rather than expanding the feasible solution space, the mitigation toolbox served a different critical function: it acted as a diagnostic probe that revealed the true depth and nature of the challenge. It proved that for this use case, achieving a governable model requires more than applying standard, off-the-shelf mitigation.

5.1.4. RQ4: what operational overhead (e.g., explanation latency, governance latency) and explainability coverage are incurred by policy enforcement?

The operational costs were negligible in the evaluated experimental

---

Box 1
Example Governance Audit Report for Institutional Review

Policy Tier: Medium_realistic; Evaluated Configuration: Logistic Regression + Threshold Optimizer (intersectional); Gatekeeper Decision: BLOCKED.

Primary Block Reasons.

Intersectional Equalized Odds gap: 0.23; *(Policy maximum = 0.12; violation confirmed under CI-aware enforcement)*; Dropout Recall: 0.59; *(Policy minimum = 0.70)*

Secondary Observations (Non-Blocking)

Calibration: Within limits (ECE = 0.04); explainability: 100% SHAP coverage; P95 explanation latency = 12 ms; Governance Overhead: P95 audit latency = 23.5 ms.

Diagnostic Interpretation (Governance-Level)

1. Disparities linked to socio-economic status (imd_band) and its intersection with gender constitute the dominant equity risk. 2. Fairness mitigation improved parity but caused unacceptable loss in recall for at-risk learners. 3. Calibration reliability and operational feasibility are not binding constraints for this configuration.

Recommended Remedial Actions (Illustrative)

1. Consider upstream data or process interventions targeting socio-economic bias. 2. Re-evaluate institutional policy thresholds or acceptable trade-offs. 3. Post-mitigation recalibration is not advised, as it invalidates enforced fairness constraints.

Audit Integrity.

1. Policy Definition: Machine-readable PoC (SHA-256 fingerprinted) 2. Artifacts Verified: Preprocessing, metrics, explainability, and diagnostics 3. Decision Mode: Conservative, CI-aware enforcement.

setting. The framework confirmed 100% SHAP coverage, and the 95th percentile SHAP latency was approximately 0.012 ms. The 95th percentile governance logging latency was 23.5 ms. These figures are well below their respective policy thresholds. A crucial finding is that explainability and auditability are not the bottlenecks; the substantive trade-offs are. We empirically show that rigorous, auditable oversight can be technically lightweight and operationally feasible.

5.1.5. RQ5: how do diagnostic outputs (shortfalls, trade-off views) inform policy tuning and adoption decisions?

This represents a core technical contribution of the proposed framework. The diagnostics convert gatekeeping from a simple pass/fail gate into an institutional learning system. By generating a Shortfall Heatmap and Policy-Tuning Suggestions, the framework provides a quantitative, multi-dimensional cost of compliance. Instead of just failing, the system proves why and by how much. For example, the diagnostics revealed:

1. **Specific Failure Modes:** The LR (TO) model was blocked by Recall, while the LR (EG) model was blocked by its Intersectional Gap and secondary ECE failure.
2. **Quantified True Challenge:** The most critical diagnostic was identifying the Intersectional Gap as the most stubborn challenge, requiring massive, unjustifiable policy relaxations of +83% to +139% to achieve compliance.

This transforms the institutional conversation. The question is no longer *"is the model fair?"* but rather: *"Our audit proves this model is blocked. To approve it, are we, as an institution, willing to accept a 139% larger fairness gap for our most vulnerable subgroups?"* This is the productive, evidence-based, and auditable output that moves governance from theory to practice.

These findings suggest three key points within the evaluated setting: (i) Our ex-ante, executable governance framework is feasible, efficient, and audit ready. (ii) The principal challenge to deployment is substantive, originating from severe, interlocking trade-offs between fairness, calibration, and performance. (iii) Our diagnostic gatekeeping provides quantified, auditable evidence to justify model rejection *or* navigate the precise, multi-dimensional costs of any policy relaxation. We now translate these insights into concrete implications.

5.2. Implications for higher education

The validation of this governance framework carries significant implications, shifting institutional AI from reactive problem-solving toward proactive, verifiable governance. Unlike previous research reliant on aspirational ethics or post-hoc audits, this framework shows that enforceable, auditable policies can be integrated ex-ante into the AI lifecycle. From an educational perspective, this shift reframes learning analytics systems from purely predictive tools into governed decision-support instruments that directly shape educational intervention, learner support, and institutional accountability.

5.2.1. For institutional leaders

This study shows that high-level ethical principles are necessary but insufficient. To avoid bias, reputational harm, and regulatory non-compliance, leaders must invest in infrastructures that make governance executable and auditable. Our framework provides such a blueprint, moving accountability from rhetoric to an engineered reality. The key implication is the transformation of evidence. When a model is blocked, leaders are no longer left with a simple failure verdict. They are instead equipped with a quantitative, diagnostic audit trail. They can now lead an evidence-based discussion, using the tuning suggestions (Table 10) to ask: *"Are we willing to accept a +139% increase in our intersectional fairness gap to deploy this model? Or does this evidence justify investing in a more fundamental (and costly) data-level intervention?"* This supports governance-aligned decision-making in educational institutions, where deployment choices must be defensible not only technically but also educationally and ethically.

5.2.2. For data science and analytics teams

This framework redefines the objective for data science teams. Their goal is no longer simply to explain the logic of a model but to prove its compliance. Our SHAP analysis showed that max_day (last interaction) was the dominant predictor, an intuitive and behaviorally sound finding. However, it also showed that sensitive attributes such as gender and highest_education were still influential. A traditional approach would stop interpretability, but our framework supports the next educationally relevant step: quantifying whether feature influence translates into non-compliant and institutionally unacceptable biases in educational outcomes. The implication is a shift in productivity. Instead of endlessly iterating mitigation techniques, the objective of data team becomes providing a quantified audit of why compliance is unachievable. When the diagnostic report proves that the most stubborn challenge is a +139% intersectional gap, the team is liberated from the "try another mitigation" churn. They now have evidence to stop turning into a probably non-compliant model, focusing on more fundamental solutions, such as fairness-aware feature engineering or alternative architecture and providing leadership with clear, data-driven justification for why the problem is deeper than simple model-level bias.

5.2.3. For educators and students

The ultimate test of any educational AI is trust. For early-warning systems to be credible, they must be demonstrably fair. By ensuring every model is validated against auditable policies, this framework provides educators with verifiable assurances of equity. This aligns formative assessment principles, where predictive signals are intended to support timely, equitable intervention rather than deterministic labeling or punitive action. By ensuring that every model is validated against auditable governance policies before deployment, this framework provides educators with verifiable assurances that analytic tools meet institutional equity standards. This, in turn, gives students confidence that algorithmically informed interventions are not systematically disadvantaged to specific demographic or socio-economic groups. Importantly, the framework prevents educationally harmful models from entering practice, rather than attempting to justify them after deployment.

5.2.4. From block verdict to a productive diagnostic

The finding of framework of zero compliant models should not be interpreted as a dead end. It is a vital and productive diagnostic that directs institutional effort. The consistent failure of the evaluated standard models and mitigations provides a clear mandate for a more fundamental approach. The path to compliance is not simply to relax policy thresholds. The path is to use the specific, quantified diagnostic outputs, such as the persistent intersectional gaps and the performance-calibration trade-offs, to facilitate a human-in-the-loop re-evaluation of the entire process. In this way, governance transforms from a final checkpoint into a catalyst for a more thoughtful, equitable, and evidence-based development lifecycle.

5.2.5. Human-in-the-loop policy formation

A critical aspect of executable governance is the translation of educational intent into measurable, enforceable thresholds. While this PoC defined quantitative policies empirically, an institutional deployment would require participatory governance. Defining acceptable levels for recall, calibration error, or fairness gap is not solely a technical matter; it also involves educational and ethical considerations. In practice, universities would convene cross-functional committees that include program directors, instructors, student representatives, data scientists, and ethics-board members. These stakeholders would review diagnostic evidence, consult regulatory frameworks such as the EU AI

Act (2024) and the NIST AI Risk Management Framework (2023), and align threshold settings with institutional missions and learner-support priorities. For example, a recall threshold of 0.70 for dropout prediction may reflect a student-safety first policy, emphasizing the prevention of false negatives in early-warning systems.

The PaC structure operationalizes these collective judgments. Once the institutionally agreed limits are encoded into JSON policy files, they become machine-enforceable, auditable, and version-controlled. This transforms educational governance from informal oversight into an explicit, reproducible component of the learning analytics lifecycle, aligning AI deployment with educational values and institutional responsibility.

5.3. Limitations of the study

While this study validates a robust, end-to-end governance pipeline, several limitations define the scope of our conclusions:

- **Dataset and Context:** This validation used only one large benchmark dataset from a distance-learning institution. It remains unclear whether these fairness trade-offs and policy thresholds apply to other settings, such as residential universities or different national regulations.
- **Metric and Outcome Scope:** The framework was focused on widely adopted observational fairness metrics (DP/EO) and a binary dropout prediction. This scope does not capture the causal mechanisms of bias or the complexities of regression/multi-class tasks, which are necessary extensions for other educational use cases.
- **Infeasibility of Standard Mitigation:** A key finding is also a practical limitation. Our framework proved that standard, off-the-shelf mitigation techniques (TO, EG) were insufficient to solve the deep, multi-dimensional trade-offs, particularly the massive intersectional fairness gaps. The framework, as a diagnostic engine, can quantify these failures but does not, by itself, create a compliant model.
- **Calibration-Mitigation Coupling:** The study does not iterate post-hoc calibration after fairness mitigation. While standard calibration techniques (Platt scaling and isotonic regression) were applied before mitigation, re-applying calibration after TO or EO would alter score distributions and invalidate enforced fairness constraints, requiring mitigation to be re-run. This study therefore treats calibration regression under mitigation as a structural trade-off rather than a remediable post-processing artifact.
- **Intersectional Sample Size Effects:** Although a minimum subgroup size threshold ($n \geq 30$) was enforced, some intersectional groups remain comparatively small relative to majority populations. While bootstrap CIs partially mitigate estimation uncertainty, fairness metrics for smaller subgroups should be interpreted conservatively. This study therefore prioritizes identifying directional and worst-case biases rather than making fine-grained claims about marginal subgroup differences.
- **Stakeholder Co-Design:** This PoC did not include formal stakeholder consultation or participatory policy co-design. The quantitative policy tiers were defined empirically to highlight technical feasibility, but their final institutional instantiation would require collaboration with educational leaders, instructors, and ethics boards.
- **Explainability Utility and Human Interpretation:** While the framework supports explanation coverage and latency as operational governance constraints, it does not quantitatively evaluate explanation quality in terms of fidelity beyond theoretical guarantees, stability under perturbation, or human interpretability. No user study or expert evaluation was conducted to assess how explanations are understood or acted upon by advisors or educators. This reflects a deliberate choice: explanations are treated as auditable artifacts for accountability rather than optimized decision-support tools.

5.4. Future research directions

These limitations define a clear and productive research agenda focused on moving from diagnosing problems to actively solving them.

- **Developing Governance-Aware Models:** The most critical future direction is to extend the diagnostic gatekeeper into a governance-aware modelling engine. This involves using the multi-dimensional shortfalls, such as the co-occurrence of ECE and fairness gaps, to drive model optimization directly. For example, future experiments could test a prototype objective function that integrates policy constraints into model training:

$$\mathscr{L}_{total} = \mathscr{L}_{perf} + \lambda_1 \cdot \text{FairnessGap} + \lambda_2 \cdot \text{ECE}$$

- where λ_1 and λ_2 encode the tolerance of the institution for inequity and miscalibration. This would show how diagnostic outputs can be fed back into the learning objective, addressing the current diagnosis and stop endpoint. The broader goal is to build adaptive models that continuously tune themselves within the boundaries of approved governance policies, ultimately transforming compliance from an external checkpoint into an embedded design principle.
- **Cross-Institutional and Real-World Validation:** A high priority is to deploy the framework across diverse institutional datasets to test the portability of the policy schemas. This must be combined with integration into live systems to study the impact of real-world consent processes (rather than simulated consent) on model fairness and representation.
- **Federated Governance for Privacy:** A key technical direction is adapting the framework for federated learning (FL). A central gatekeeper in a federated system can provide governance and check for bias across separate data sources, enabling organizations to work together on building and validating reliable models, all while keeping their own data private and under local control.
- **Closing the Human-in-the-Loop:** Future work must study how stakeholders use the diagnostic reports in practice. Evaluating how leadership, ethics boards, and data teams collaboratively use the tuning suggestions to deliberate and set policy will be the ultimate test of moving this from a technical pipeline to a fully operational, human-centered governance system.

These directions would extend the framework from a single-institution PoC into a scalable, federated, and participatory governance model capable of supporting trustworthy educational AI across diverse contexts.

6. Conclusion

Higher education is increasingly shaped by artificial intelligence (AI) systems whose decisions affect student support, progression, and institutional accountability. However, a persistent gap remains between aspirational ethical principles and mechanisms that can enforce, audit, and justify these principles in practice. This study addresses that gap by showing an end-to-end Policy-as-Code (PaC) governance framework that translates institutional requirements into executable, machine-verifiable rules integrated directly into the model evaluation and deployment pipeline. Using a dropout-risk prediction task on the OULAD dataset as a proof of concept (PoC), the empirical results show that, under the evaluated dataset, policy tiers, and model-mitigation configurations, none of the sixteen candidate models achieved full compliance across all governance dimensions. Rather than treating this outcome as a failure, the framework exposes it as a productive diagnostic signal. The results indicate that the dominant challenges to compliance in this setting arise from substantive trade-offs among fairness, calibration, and predictive performance, rather than from technical or operational constraints such as latency, explainability cost, or audit overhead. The principal contribution of this study is therefore not a new predictive model, but a diagnostic and auditable governance blueprint. By coupling executable policies with confidence-interval (CI)-aware gatekeeping, multi-strategy mitigation analysis, and a tamper-evident audit chain, the framework produces quantitative, explainable evidence for every approval or rejection decision. This capability transforms governance from a retrospective compliance check into an active decision-support process, enabling institutions to understand why models fail, where trade-offs arise, and what policy relaxations would be required for deployment. Importantly, the findings do not claim general infeasibility of fair or trustworthy educational AI. The findings indicate that meeting all governance requirements at once in this context is complex and demands thoughtful institutional action beyond standard mitigation methods. The framework explicitly supports this deliberation through human-in-the-loop policy formation, allowing educators, data scientists, and ethics boards to engage in transparent, evidence-based decision-making grounded in auditable diagnostics. This study shows that trustworthy educational AI is not achieved by lowering ethical standards, but by operationalizing them within computational design. The proposed PaC framework provides a reproducible, diagnostically rich foundation for governance-aware AI systems, shifting institutional trust requirements from external checkpoints to internal design constraints. While validated here as a PoC, the framework offers a scalable direction for future governance-aware modelling efforts and cross-institutional deployments.

CRediT authorship contribution statement

Edmund Evangelista: Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Syed M. Salman Bukhari:** Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation.

Ethics approval and consent to participate

This study analysed a publicly available, fully de-identified educational dataset (the Open University Learning Analytics Dataset, OULAD). No new data were collected and no direct interaction with human participants occurred.

Data availability statement

The Open University Learning Analytics Dataset (OULAD) analysed during the current study is publicly available from the UK Open University's Knowledge Media Institute at: https://analyse.kmi.open.ac.uk/open_dataset.

Funding

This research was funded by Zayed University through the Research Incentive Fund (RIF), grant number R22013.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgements

This research was supported by the Research Incentive Fund (RIF) Award Grant Activity Code R22013 from Zayed University, United Arab Emirates.

References

Agarwal, A., Beygelzimer, A., Dudík, M., Langford, J., & Wallach, H. (2018). A reductions approach to fair classification. In *Proc. int. conf. machine learning (ICML '18), 80* pp. 60–69). PMLR.

Baker, R. S. J.d., & Siemens, G. (2012). Learning analytics and educational data mining: Towards communication and collaboration. In *Proc. 2nd int. Conf. Learning analytics & Knowledge (LAK '12)* (pp. 252–254). https://doi.org/10.1145/2330601.2330661

Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness through awareness. In *Proc. Innovations in theoretical computer science conf* (pp. 214–226). ITCS '12). https://doi.org/10.1145/2090236.2090255.

Guo, M., Pleiss, Y., Sun, S., Weinberger, F., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proc. int. conf. machine learning (ICML '17), 70* pp. 1321–1330). PMLR.

Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning.* Boston, MA: Center for Curriculum Redesign.

Holmes, W., Porayska-Pomsta, K., Holstein, K., Sutherland, E., Baker, T., & Russell, J. (2021). Ethics of AI in education: Towards a community-wide framework. *British Journal of Educational Technology, 52*(4), 1647–1667. https://doi.org/10.1111/bjet.13102

Kearns, M., Neel, S., Roth, A., & Wu, Z. S. (2018). Preventing fairness gerrymandering: Auditing and learning for subgroup fairness. In *Proc. int. conf. machine learning (ICML '18), 80* pp. 2569–2577). PMLR.

Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open university learning analytics dataset. *Scientific Data, 4*, Article 170171. https://doi.org/10.1038/sdata.2017.171

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in neural information processing systems, 30* pp. 4765–4774). NeurIPS '17).

Ma, J., Zhao, Z., Xie, Z., Zhang, Y., & Zhou, G. (2025). LE-DLCM: Decoupled learner and course modeling with large language models for enhanced course recommendation. *Knowledge-Based Systems*, Article 115135.

Macfadyen, L. P., & Dawson, S. (2010). Mining LMS data to develop an 'Early Warning System' for educators: A proof of concept. *Computers & Education, 54*(2), 588–599. https://doi.org/10.1016/j.compedu.2009.09.008

Mökander, J., Axente, M., Casolari, F., Floridi, L., & Watson, D. (2022). Conformity assessments and post-market monitoring: A guide to AI regulation. *Minds and Machines, 32*(4), 495–520. https://doi.org/10.1007/s11023-022-09617-2

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0).* Gaithersburg, MD: U.S. Dept. Commerce.

Plečko, D., & Bareinboim, E. (2024). Causal fairness analysis: A causal toolkit for fair machine learning, found. *Trends Machine Learning, 17*(3), 304–589. https://doi.org/10.1561/2200000106

Raji, I. D., Smart, A., White, R., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proc. Conf. Fairness, accountability, and transparency (FAccT '20)* (pp. 33–44). https://doi.org/10.1145/3351095.3372873

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier. In *Proc. 22nd ACM SIGKDD int. Conf. Knowledge discovery & data mining* (pp. 1135–1144). KDD '16). https://doi.org/10.1145/2939672.2939778.

Selwyn, N. (2019). *Should robots replace teachers? AI and the future of education.* Cambridge, UK: Polity Press.

Slade, S., & Prinsloo, P. (2016). Learning analytics ethical dilemmas: A review of current challenges and emerging tensions. *Journal of Learning Analytics, 3*(1), 64–79. https://doi.org/10.18608/jla.2016.31.5

Smuha, N. A. (2025). Regulation 2024/1689 of the Eur. Parl. & Council of June 13, 2024 (eu artificial intelligence act). *International Legal Materials, 64*(5), 1234–1381.

United Nations Interregional Crime and Justice Research Institute. (2024). *Principles for responsible AI innovation.* UNICRI.

Weerts, H., van der Meer, M. W., Bird, S., & van der Maas, M. W. (2023). Fairlearn: Assessing and improving fairness of AI systems. *Journal of Machine Learning Research, 24*, 1–7.

Williamson, B., & Piattoeva, N. (2019). Objectivity as standardisation in data-scientific educational governance: Grasping the global through the local. *Research in Education, 101*(1), 69–91. https://doi.org/10.1177/0034523719836820

Williamson, B., & Selwyn, N. (2019). Data infrastructure and education policy: The digital forecasting of achievement and failure. *Social Science Computer Review, 37*(2), 135–153. https://doi.org/10.1177/0894439318782891

Yli-Huumo, J., Ko, D., Choi, S., Park, S., & Smolander, K. (2016). Where is current research on blockchain technology? A systematic review. *PLoS One, 11*(10), Article e0163477. https://doi.org/10.1371/journal.pone.0163477